

On explanatory theories of speech categorization

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Draft version February 4, 2026

Abstract

The study of speech categorization has a long history. Two phenomenological laws of speech categorization are statable based on this extensive history: monotonicity and sigmoidality of the identification/categorization functions. While prior work has heavily focussed on accounting of observed data in terms of specific theories, and computationally modelling observed patterns, far less work has been done on formally comparing theories to study their explanatory potential, particularly their explanatory value with respect to the two laws. Taking up the latter challenge, in this paper, I formally compare four theories (or frameworks) that have been popular in the literature: (a) Categorical Boundary theories, (b) Prototype theories, (c) Bayesian theories, (d) Exemplar theories. Out of the four, only Category Boundary theory was found to have a clear and simple explanatory account for the two laws. Prototype theory is shown to be consistent with the law of monotonicity but not sigmoidality. Bayesian categorization and Exemplar theories were found to not be explanatory of either law except when severe, currently *ad hoc*, restrictions were imposed on the category distributions.

Keywords: speech categorization, monotonicity, sigmoidal curve, theories of speech categorization, formal analysis.

1 Introduction

The study of cognition, as with all of Science, strives for *explanations*. In the literature, there are at least four different paths researchers have trodden in search of explanations of cognitive phenomena. First, experimentation and concomitant statistical modelling to obtain potentially relevant facts that could form a part of the explanation of the relevant cognitive phenomenon. Second, computational modelling of theories to generate expectations that can be compared against observed data. Third, fit computational models to observed data by adjusting free parameters. Finally, (formal/mathematical) analysis of theories in order to see if they correctly explain idealized phenomena (or “laws”). While most research has focussed on the first two, it is the last path that really allows us to see if our theories are explanatory or not (Cummins 2000). Particularly compelling early efforts in the history of modern Cognitive Science in this direction were Miller’s (1957) explanation of Zipf’s law as stemming from random processes, and Chomsky’s (1959) argument that syntactic knowledge needs more expressive formalisms than finite-state models. A crucial aspect of explanatory theories is that they *entail* the relevant set of (idealized) laws (Hempel 1942; Hempel and Oppenheim 1948; Kitcher 1989). Equally important in seeing what is a true explanation is to ensure that the counter-factual does not occur (Van Fraassen 1980).¹ Consequently, the assessment of the explanatory value of a theory requires two things: (a) a statement of what the relevant phenomenological/observational laws are, (b) a formal analysis of the space of possibilities for the theory to see if it entails the laws. This will be the analytic conceit in this paper in order to assess the explanatory value of different theories of speech categorization.

Categories and categorization have been at the heart of the cognitive enterprise right from its inception in the 1950’s (see Cohen and Lefebvre 2005; Harnad 1987, for discussion). A particularly fruitful avenue of research has been on the categorization of color continua in vision (Bornstein 1987) and categorisation of speech sounds in speech perception (Apfelbaum et al. 2022; Fry et al. 1962; Kuhl 1991; Liberman et al. 1967, 1957; Pisoni 1973; Pisoni and Lazarus 1974; Toscano et al. 2010). In the realm of speech perception, such work came to prominence in the modern era with the foundational work of Liberman et al. (1957). While the work gained fame for claims related to the theory of **Categorical Perception**, their work included a crucial

¹Note, this is a distinction that has recently come to fore in debates about whether large language models are truly explanatory theories or just data fitting accounts (Chomsky and Moro 2022; Chomsky et al. 2023; Piantadosi 2023).

observation that average (across-listener) identification curves of continua in a 2-category identification task have a specific shape — they are sigmoidal as schematized in Figure 1. It is this observation, which has since been observed repeatedly in the literature (Hessen and Schouten 1999; Massaro and Cohen 1983; McMurray 2022, amongst others) that forms the basis of the current paper. Notably, similar observations of sigmoidality have been made about color categorization in the visual perception literature (Beare 1963; Bornstein and Korda 1985; Bornstein 1987; Huette 2012). Similar sigmoidal curves have also been observed in historical language change (Kroch 1989; Labov 2001), language acquisition (Yang 2001), and phonotactics (Hayes 2022), amongst other fields.

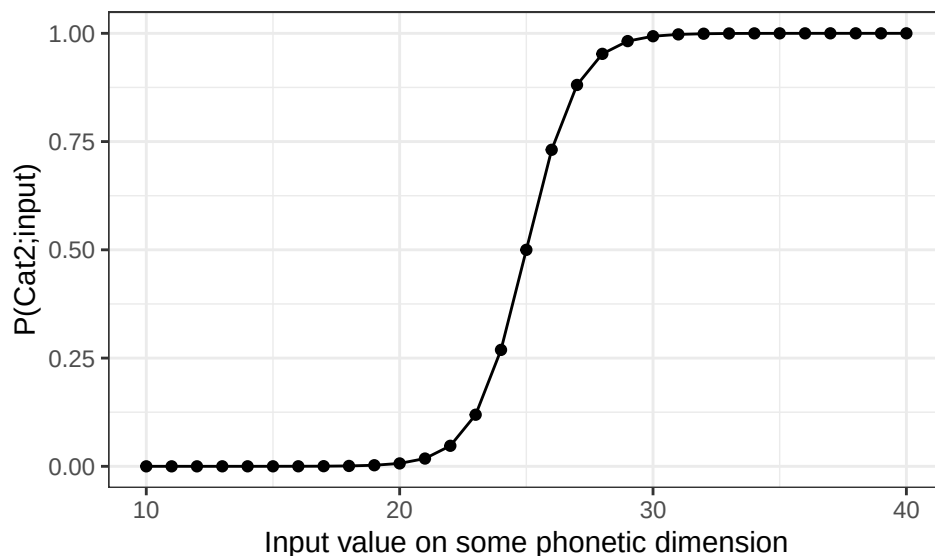


Figure 1: An example of a sigmoidal identification function for a continuum of stimuli

Though early studies observed that the phenomenon of categorical perception was weaker with vowels than with consonants (Fry et al. 1962), later research has argued that the failure to observe the phenomenon with vowels might be due to details related to experiment design (Hessen and Schouten 1999; Pisoni 1975; Pisoni and Lazarus 1974). For our purposes, even the early study of Fry et al. (1962) showed that the identification function was sigmoidal, albeit perhaps a bit flatter than is the case with consonants. But as McMurray (2022) correctly points out, the flatness/steepness of a curve cannot be established without fixing the endpoints of the continua to reflect identical ranges for each continuum. Given vowel and consonant identification experiments typically study different sets of phonetic dimensions, the relative flatness/sigmoidalness

of the identification curves is difficult to establish in a more precise (non-impressionistic) way.

It should be noted at the outset that the observation of sigmoidality of the identification function in a 2-category identification case is often based on an impressionistic appraisal. In order to make progress on the issue, the observation can be broken down into “weak” and “strong” laws. The weak form of the law (1a) states that the (2-category) identification function changes monotonically; in this form, the law makes claims only about the direction of change, and not the shape of the change. A similar observation was made by Shepard (1962a,b) in relation to work on Multidimensional scaling. In its stronger form (1b), the law makes specific claims about both the direction of the change and the shape of the categorization function.

1. Two phenomenological laws of speech categorization (2-category case)

(a) **Weak law:** The categorization function increases/decreases monotonically increasing along a phonetic dimension.

(b) **Strong law:** The categorization function has a sigmoidal shape along a phonetic dimension.

These laws are so well established that they have been used to understand: (a) how the perception of continua changes with repeated presentation — a phenomenon titled “selective adaptation” (Eimas et al. 1973; Eimas and Corbit 1973); (b) how the perception of stimuli along a continuum changes when an ambiguous token is disambiguated based on lexical information (Caplan et al. 2021; Eisner and McQueen 2005; Kraljic and Samuel 2005; Nelson and Durvasula 2021; Norris et al. 2003) or visual information (Bertelson et al. 2003; Van Linden and Vroomen 2007; Vroomen et al. 2004) — a phenomenon called “perceptual retuning” or “perceptual learning”. It is also worth noting that if the weak law (1a) were violated, then it opens the door to the possibility of “non-local” categorization, wherein categories have disjoint and non-contiguous ranges along a single phonetic dimension. Crucially, such non-locality is never observed naturally for speech categories (Heffner et al. 2019).

Importantly, both the laws in (1) are *phenomenological laws* (Cartwright 1983), and are related to average response rates of listeners. Furthermore, phenomenological laws are about *observations* and not underlying theoretical constructs, and it would be a mistake to infer the underlying (theoretical) computations directly from such laws, *i.e.*, such phenomenological laws are ultimately epiphenomenal and derived from more

abstract underlying theoretical constructs. More directly relevant to identification functions, Estes (1956) and Sidman (1952) noted many decades ago that the same average response curve is consistent with a multitude of underlying computations or theories of such computations. While such ambiguity necessarily exists, Estes (1956, p. 134) also correctly points out that such an average response curve “does provide a means of testing exact hypotheses about them” (*i.e.*, underlying computations). Therefore, the laws can be used as the idealised empirical bases that can be used to probe the explanatory value of different theories of speech categorization (as discussed by Kitcher 1989, more generally for scientific explanations). Before proceeding to discuss the theories that are relevant for the current paper, it is worth addressing two issues with the above framing of the laws in (1). First, one could object that the 2-category case is a very specific instance of a more general issue of n -category distinctions along a phonetic dimension. However, it is straightforward to re-phrase the laws in (1) in terms of relative support between any two categories — such functions will still appear to observe the two laws. For the rest of the paper, I will lay aside this complication, and focus on the 2-category case for the sake of both expository and analytic convenience.

Second, as alluded to above, there has been a long debate of the nature of the initial percept, whether it is categorical or gradient (Caplan and Durvasula [under review](#); Fry et al. 1962; Hessen and Schouten 1999; Liberman et al. 1967, 1957; Massaro and Cohen 1983; McMurray 2022, amongst others). While researchers disagree on the nature of the initial percept, there is agreement that categorization must happen at some level to interface with other aspects of language and cognition. The focus of the current paper is on the narrower issue of theories of speech categorization.² Specifically, I will formally compare the predictions of four speech categorization theories with respect to the shape of the identification function (2). The order doesn’t reflect the chronological order of development of these ideas — both Bayesian (Miller and Gelman 2020) and Exemplar (Silverman 2012) theories can be traced back to at least the 19th century. Instead, the order of presentation relates to how successful (most to least) such theories are in predicting/explaining the laws of speech categorization stated in (1). Note, throughout, I will refer to each of the four theories in the plural (“theories”), as the basic formal idea can be couched in different specific instantiations of the theories, *i.e.*, I use the plural “theories” to refer to specific general “frameworks”, in order to keep the terminology

²Of course, if the initial perceptual representation itself is in terms of discrete categories, then the result applies to general (early) speech perception too (see Caplan and Durvasula [under review](#), for recent discussion).

minimal.

2. Theories of speech categorization

(a) Category Boundary theories

(b) Prototype theories

(c) Bayesian theories

(d) Exemplar theories

In what follows, I present a brief overview of the four different theories from a bird-eye view, and develop much more formal characterizations of each of them in the following section. First, Category Boundary theories are implied in the foundational work on speech perception and categorization in the modern era (Fry et al. 1962; Liberman et al. 1957). In such research, categories were thought to be identified based on specific thresholds employed on phonetic dimensions. In fact, for many researchers in that era, category exponents were thought to have discontinuities in each relevant phonetic dimension, such that the discontinuities could be used as threshold to identify categories (Rosch 1978). Since the early work, such category boundary based categorization theories have been argued to be inconsistent with observed human behavior (Massaro and Cohen 1983; McMurray 2022; McMurray et al. 2008). However, such arguments have an unstated assumption that the category boundary has no associated variance/noise. As will be shown in this paper, once we acknowledge that listeners' categorical boundaries have associated noise/variance, such criticism cannot be sustained (see Huette (2012) for a similar observation about sigmoidality in color perception).

Second, Prototype theories involve speech perception and categorization as a comparison to *prototypes*, which are most representative of a category and least representative of other categories (Rosch 1977, 1978; Rosch and Mervis 1975, amongst others). While it is natural to consider prototypes to be the representations of the corresponding categories, Rosch (1978, p. 40) explicitly disavows the necessity of this connection and suggests that “such a statement remains an unspecified formula until it is made concrete by inclusion in some specific theory of representation”. Finally, it is worth noting that, while fuzzy logic based theories of speech perception were primarily introduced to integrate multiple sources of information, the crucial computation relevant to the identification function is in terms of goodness of match between the input stimulus and

prototype descriptions (Massaro and Oden 1980; Massaro and Simpson 1987; Oden 1981; Oden 1984; Oden and Massaro 1978). Consequently, the formalization and results discussed for prototype theories will also apply to such theories.

Third, Bayesian theories of perception and categorization involve the use of Bayes theorem to reverse infer the best category given the input, parametric (typically, Normal³) distributions of the categories, and the priors on the categories (Feldman 2015; Kersten and Yuille 2003; Norris and McQueen 2008; Tenenbaum and Griffiths 2001; Wolpert and Ghahramani 2005). Bayesian theories have been quite popular in recent times and have been used to account for a variety of phenomena: the perceptual magnet effect (Feldman and Griffiths 2007; Feldman et al. 2009; Kronrod et al. 2016), selective adaptation and perceptual retuning (Kleinschmidt and Jaeger 2015; Tan and Jaeger 2025; Xie et al. 2023), about co-articulatory compenzation (Sonderegger and Yu 2010; Yu 2011), amongst others. Such theories have often been argued to be computational-level theories in the sense of Marr (1982), a topic that we will return to later in the paper.

Fourth, Exemplar theories conceive of stored representations as high-dimensional (likely, 1000's of phonetic dimensions according to Port (2007)) episodic memories of each instance of experience related to a category. In speech perception, such episodic instances are used, weighted by their frequency and memory decay effects, in consort with specific biases for different categories. While exemplar theories of psychological theories of speech are much older than some of the theories considered (see Du and Durvasula 2025; Silverman 2012, for discussion of these issues), precise theories of speech perception and categorization became more popular after Nosofsky's (1987, 1988a,b) foundational work. Since then, there has been an incredibly rich array of work arguing for such representations in speech perception (Goldinger 1996, 1997, 1998; Johnson 1997; Pierrehumbert 2016, amongst others).

There are two points worth highlighting regarding the above four theories. First, all the above theories are theories of speech categorization, but don't necessarily commit to precise learning mechanisms. However, issues of how such knowledge (categories, category distributions, categorical boundaries, prototype information, priors, weights, ...) is learnt by human beings are independent of what and how such knowledge is utilised in speech categorization; and, we shall limit ourselves to the latter issues in the paper.

³Throughout, I will stick to the term Normal, instead of Gaussian, with no particular reasoning except consistency.

Second, it is to be noted that the variance or noise in categorization enters in different ways in each of the different theories. In Category Boundary theories, noise in the categorization can potentially be attributed to variance in the category boundary itself. In Prototype theories, noise in categorization can be attributed to variance or noise in the distance estimation function. In Bayesian theories, the noise is inherent to the categories themselves in that they have an associate variance for each phonetic dimension. And in Exemplar theories, the noise in categorization could be attributed to the inherent variation in the stored exemplars or to distance estimations from such exemplars to input stimuli. So, an element of formally comparing the different theories relates to what are the appropriate theoretical constructs to which variances/noise should be linked.

In what follows, I present a formal description of each of the theories and then derive specific characteristics of the identification function through formal proofs and simulations (Section 2). The pay off of developing formal theories (as needed) is that we are able to derive precise predictions about the identification functions, and the conditions under which such functions predict/explain the laws in (1). Furthermore, the formalization allows us to derive a host of other phenomena related to the shapes of the identification functions. Finally, in Section 3, I will conclude with brief remarks about further consequences of the analytic results, and about possible ways to extend this line of research.

2 Formal models of speech categorization

Our objective in this paper is to see if the weak and strong laws of speech categorization are explained by each of the theories studies. This requires formalizing and assessing each of the theories. Before we proceed to formally modelling and assessing the explanatory value of each theory, it is important to make the formal assumptions about the nature of noise/variation involved explicit.

Throughout, I will assume that the Normal distribution is a reasonable approximation of the noise/variance distribution, as has been argued both in nature more generally (Lyon 2014) and in psychophysics mode specifically (Guilford 1954; Thurstone 1927; Treisman 1999).⁴

Furthermore, I will also focus on a single phonetic dimension in a 2-category system. This simplifying

⁴But see Limpert et al. (2001) who argue that perhaps log-normal distributions are better approximations.

assumption makes the formal analysis easier to present. Furthermore, if the k-phonetic dimensions (in reality) are independent in the underlying system, then there is no loss of generality in the results.

2.1 Category-boundary based categorization

As per Category Boundary theories, a specific category boundary results in a discrete category output to a specific input.⁵ As a first approximation, one might assume that the category boundary falls at the singular point where the densities of the distributions are identical — this could potentially be learned incrementally based on trying to minimize error rates of category identification.

Without loss of generality, let's assume Cat_1 is the category with the lower values, and Cat_2 is the category with the higher values. If an input value (x) is higher than the category boundary (θ), then the input will be categorized as Cat_2 , and if it is lower than the boundary, then it is categorized as Cat_1 . So, the category boundary acts as an indicator function which maps an input value to either 0 or 1. Therefore, we can formalize it as follows, where the value of 1 represents Cat_2 (the category with the higher values) and 0 represents Cat_1 (the category with the lower values).

$$I : X \rightarrow \{0, 1\} \quad (1)$$

$$I(x) = \begin{cases} 1 (Cat_2) & \theta \leq x \\ 0 (Cat_1) & \theta > x \end{cases} \quad (2)$$

However, as mentioned, the category boundary itself has a distribution, say a normal distribution with a mean (μ_θ) and variance (σ_θ^2).

Before we derive the relevant probability curve, it is helpful to use an example. In Figure 2 below, the θ distribution is represented with a Normal curve, and the value of the input is marked. If the value of the θ drawn from the distribution in any single instance is below x , then the indicator function, I , will output 1, indicating that Cat_2 is perceived. In fact, in the example below, if the μ_θ is drawn as the random value of the categorical boundary, then it is easy to see that the system should output Cat_2 as the response.

⁵A formal result very similar to the current one is presented by Caplan and Durvasula ([under review](#)).

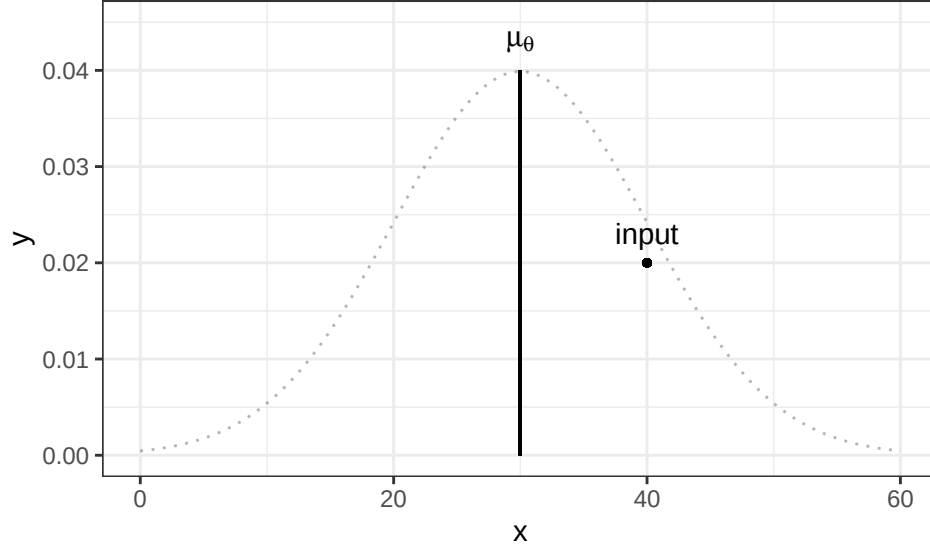


Figure 2: Distribution of category boundary values in comparison to the input.

Now, we can ask, what is the probability of perceiving *Cat*₂ when the input is *x*? We know that

$$\mathbb{P}(\text{Cat}_2; x) = \mathbb{P}(I = 1; x) \quad (3)$$

And, in turn, because the probability of the indicator function returning 1 is equal to the probability that the categorical boundary is less than *x*,

$$\mathbb{P}(I = 1; x) = \mathbb{P}(\theta \leq x) \quad (4)$$

The above is of course just the cumulative distribution function (c.d.f.) for a Normal distribution, with a mean (μ_θ) and variance (σ_θ^2). Therefore,⁶

$$\mathbb{P}(\text{Cat}_2; x) = \Phi\left(\frac{x - \mu_\theta}{\sigma_\theta}\right) \quad (5)$$

Since the c.d.f. of a Normal distribution is always a sigmoidal function for any combination of values of mean and variance parameters, from the above, we derive the fact that the identification function for a category-boundary based perceptual system *must* be sigmoidal irrespective of the parameters involved.

⁶Note, the same conditional probability below can also be derived as an expectation of the indicator function (I) given the input (*x*), akin to the calculation of a Bernoulli trial, albeit with a probability distribution function, instead of a mass function.

201 Finally, $\mathbb{P}(I = 1; x) + \mathbb{P}(I = 0; x) = 1$; therefore,

$$\mathbb{P}(Cat_1; x) = \mathbb{P}(I = 0; x) = 1 - \mathbb{P}(I = 1; x) = 1 - \Phi\left(\frac{x - \mu_\theta}{\sigma_\theta}\right) \quad (6)$$

202 An interesting aspect of Eqs. 5-6 is that as the variance increases, the identification curve straightens out;
 203 and in the limit, the identification curve is expected to be linear and monotonically increasing/decreasing
 204 with the input stimulus. So, the formalization allows us to account for both apparently sigmoidal and
 205 apparently more linear identification curves observed in the literature as a consequence of difference in the
 206 variance of the category boundary.

207 Let's simulate the expected identification function for two categories with specific category boundary
 208 values in one phonetic dimension. Let's say the first categorical boundary is $\theta_1 \sim N(\mu_\theta = 15, \sigma_\theta^2 = 5)$, and a
 209 second categorical boundary is $\theta_2 \sim N(\mu_\theta = 15, \sigma_\theta^2 = 60)$. We can now repeatedly simulate the comparison
 210 of input values along a continuum against specific draws from the category boundary distribution (say, 1000
 211 times for each input value) and compare the observed identification function against the predicted probability
 212 from the above analytic result. As can be seen in Figure 3, the identification function looks more sigmoidal
 213 on the left with lower variance in the categorical boundary, and looks flatter in the same region with increased
 214 variance in the category boundary (right). Notably, both are actually sigmoidal in abstract formal terms, as
 215 the c.d.f of the standard Normal distribution is sigmoidal.

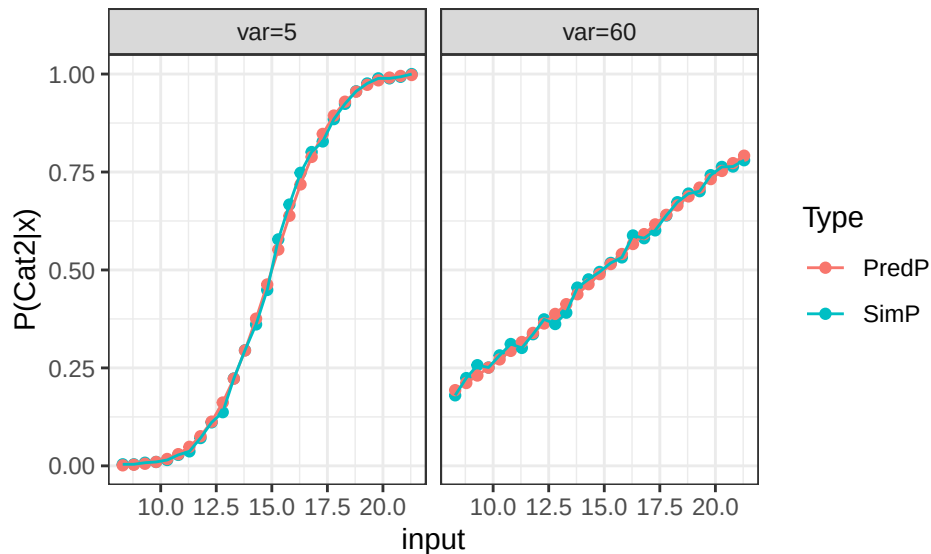


Figure 3: Identification function under Category Boundary theory for different variances associated with the boundary.

While as mentioned earlier, I have focussed on the 2-category identification case in a single dimension, the result does scale to more complex cases. For example, if there are k -dimensions, the category boundary will be a hyper-plane. Furthermore, if the distributions are independent on all of them, then the sigmoidality is predicted to occur on each of the k dimensions.

Finally, if the locus of variation of the categorical boundary is within the individual, then it follows that a single individual will show a gradient and sigmoidal curve response to a continuum despite using a categorical threshold in every instance.

2.2 Prototype theories

As per Prototype theories, a specific category only has a prototype value for each of the k phonetic dimensions. Based on Rosch (1977, 1978) and Rosch and Mervis (1975), the prototype values for continuous phonetic dimensions can be assumed to be the mean of the category. Let's say that the prototype values for each category, Cat_1 and Cat_2 , are the two vectors $\vec{\mu}_1 = (\mu_{1,1}, \mu_{1,2}, \dots, \mu_{1,k})$ and $\vec{\mu}_2 = (\mu_{2,1}, \mu_{2,2}, \dots, \mu_{2,k})$. In a single phonetic dimension, the prototypes would reduce to single values, μ_1 and μ_2 . Let's further assume that the identification of a category is based on some distance function, $D(x,y)$, which is necessarily a non-negative symmetric function (so, $D(x,y) \geq 0$; $D(x,y) = D(y,x)$).

Without loss of generality, let's assume Cat_1 is the category with the lower mean value, and Cat_2 is the category with the higher prototype (*i.e.*, mean) value, along some phonetic dimension. So, for a single phonetic dimension, if the input x is less than the mid-point between Cat_1 and Cat_2 , then it will be categorized as Cat_1 ; otherwise, it will be categorized as Cat_2 . Relatedly, if there are k phonetic dimensions, the threshold will be the hyperplane that bisects or is equidistant between the two category mean values.

The above single dimension case can be re-phrased as, if the difference in distances from the categories ($Diff(x) = D(Cat_1, x) - D(Cat_2, x)$) is above 0, then the input will be perceived as Cat_2 ; otherwise, it will be categorized as Cat_1 . We can formalize it as follows, where the value of 1 represents Cat_2 (the category with the higher values) and 0 represents Cat_1 (the category with the lower values).

$$I : X \rightarrow \{0, 1\} \quad (7)$$

$$I(x) = \begin{cases} 1 (Cat_2) & Diff(x) > 0 \\ 0 (Cat_1) & otherwise \end{cases} \quad (8)$$

So far, we have proceeded by assuming that there was no noise or variation in the distance estimate; *i.e.*, the distance estimation is perfect. Now, let's further assume that the distance function itself has a separate noise distribution ($\epsilon \sim N(\mu_\epsilon = 0, \sigma_D^2)$) that is independent and identically distributed for each distance measure. Therefore, the difference can actually be represented with a random variable for each value of input, x , which leads to the following consequences:

$$Diff(x) = (D(\mu_1, x) + \epsilon) - (D(\mu_2, x) + \epsilon)$$

$$E[Diff; x] = E[(D(\mu_1, x) + \epsilon) - (D(\mu_2, x) + \epsilon)] = D(\mu_1, x) - D(\mu_2, x)$$

$$Var[Diff; x] = Var[(D(Cat_1, x) + \epsilon) - (D(Cat_2, x) + \epsilon)] = Var[\epsilon] + Var[\epsilon] = 2\sigma_D^2$$

The expected value of the differences is simply the difference in the distances from the prototypes to the input value, while the variance in the difference measurement is $2\sigma_D^2$. And since the error (ϵ) function is

248 Normally distributed,

$$Diff(x) \sim N(E[Diff;x], Var[Diff;x]) \quad (10)$$

249 As in the previous sub-section, it is helpful to use an example. In Figure 4, the $Diff(x)$ distribution is
 250 represented with a Normal curve, $N(E[Diff;x], 2\sigma_D^2)$, and the value of 0 difference is marked with a vertical
 251 dashed line. If the value of the $Diff(x)$ in any single instance is above 0, then the indicator function, I
 252 will output 1, indicating that Cat_2 is perceived. For any input x , the difference in distance from each of the
 253 prototypes is $D(\mu_1, x) - D(\mu_2, x)$, which also happens to be the expected value of the $Diff(x)$ in noise.

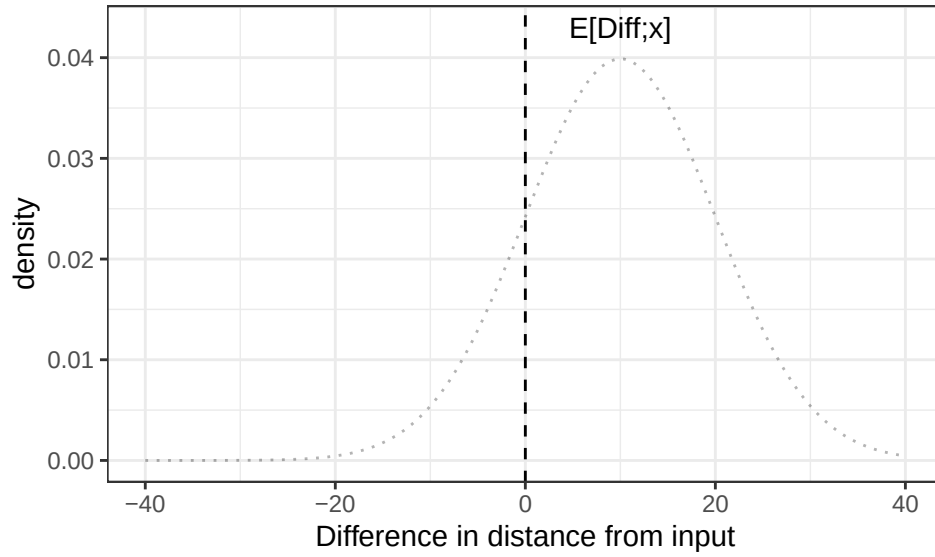


Figure 4: Difference in distances from each category prototype to the input. $E[Diff;x]$ represents the differences in distances for the input x .

254 We can now ask, what is the probability of perceiving Cat_2 when the input is x ? We know that

$$\mathbb{P}(Cat_2; x) = \mathbb{P}(I = 1; x) \quad (11)$$

255 And, in turn, because the probability of the indicator function returning 1 is equal to the probability
 256 that the difference in distances being greater than 0.

$$\mathbb{P}(I = 1; x) = \mathbb{P}(Diff(x) > 0) \quad (12)$$

However, instead of attaching the variance to the difference value and seeing if it is greater than 0, we could attach the variance to the zero threshold value itself and keep the difference, $\text{Diff}(x)$, a constant for a given x . That is, we can create a new random variable $\theta \sim N(0, 2\sigma_D^2)$ and ask if the distance is less than that random variable. Furthermore, given that we are dealing with a continuous distribution, $\theta < \text{Diff}(x)$ is the same as $\theta \leq \text{Diff}(x)$. Therefore,

$$\begin{aligned} \mathbb{P}(\text{Cat}_2; x) &= \mathbb{P}(\text{Diff}(x) > 0) \\ &= \mathbb{P}(\theta < \text{Diff}(x)), \text{ where } \theta \sim N(0, 2\sigma_D^2) \\ &= \mathbb{P}(\theta \leq \text{Diff}(x)), \text{ where } \theta \sim N(0, 2\sigma_D^2) \end{aligned} \quad (13)$$

Eq. 13 is the cumulative distribution function (c.d.f.) for a Normal distribution ($N(0, 2\sigma_D^2)$). However, at this point, it is worth noticing that in 1 dimension, the Diff function has a one-to-one correspondence with the input value *only* between the means of the two categories, but it is a constant function beyond the category means as in Eq. 14. Just to ensure the intuition is clear, it is worth working through a specific example: if there is an input x that is beyond the Cat_2 prototype (m_2), then the Euclidean distance to Cat_2 will be $x - \mu_2$. Similarly, the Euclidean distance to Cat_1 will be $x - m_1$. However, μ_2 itself can be written as $\mu_1 + C$, where C is some constant. Consequently, the difference in distances will be $(x - \mu_1) - (x - \mu_2) = (x - \mu_1) - (x - (\mu_1 + C)) = (x - \mu_1) - (x - \mu_1 - C) = C$. That is, the difference in distance is constant when the input is beyond the Cat_2 prototype value. A similar calculation will show that the difference in distance function is a constant when the input value is less than μ_1 .

$$\text{Diff}(x) = \begin{cases} \mu_1 - \mu_2 & [+ve \text{ constant}] & x > \mu_2 \\ D(\mu_2; x) - D(\mu_1; x) & & \mu_1 < x < \mu_2 \\ \mu_1 - \mu_2 & [-ve \text{ constant}] & x < \mu_1 \end{cases} \quad (14)$$

Consequently, the probability function above can be decomposed into three cases and presented in terms

273 of the standard Normal c.d.f. (Φ):

$$\mathbb{P}(Cat_1; x) = \begin{cases} \Phi\left(\frac{\mu_1 - \mu_2}{2\sigma_D^2}\right) & [constant \geq 0.5] & x > \mu_2 \\ \Phi\left(\frac{Diff(x)}{2\sigma_D^2}\right) & & \mu_1 < x < \mu_2 \\ \Phi\left(\frac{\mu_1 - \mu_2}{2\sigma_D^2}\right) & [constant \leq 0.5] & x < \mu_1 \end{cases} \quad (15)$$

274 That is, the expected identification function is equivalent to the cumulative distribution function (c.d.f.)
 275 for the Standard Normal distribution *only between the means of the two categories*, but is constant oth-
 276 erwise, *i.e.*, if the input is outside the range of the means, the probability steady-states to some constant
 277 value. Crucially, there is no non-locality in the expected identification function, because monotonicity is not
 278 violated.

279 An interesting aspect of Eq. 15 is that the steady-state values are contingent on the prototype values,
 280 and on the variance in the distance measurement. The more the variance in the distance measurement, the
 281 further away is the respective steady-state probability value from the extreme (0 or 1). In the limit, when
 282 variance is infinite, the identification function is expected to be a horizontal line, suggesting that there is no
 283 sensitivity to the categories at all.

284 It is worth noting here that when the input x is between μ_1 and μ_2 , the $E[Diff; x]$ will always be trapped
 285 between the values for $\mu_1 - \mu_2$ at the mean values. Consequently, the probability of Cat_2 identification in
 286 the range (μ_1, μ_2) will necessarily be between the steady-state values that appear beyond the mean values.
 287 So, the above function will *look* sigmoidal for sufficiently small values of error in the distance calculation
 288 (*i.e.*, σ_{Diff}^2 is sufficiently small). However, if the error is sufficiently large, then the identification function
 289 between the prototypes is expected to look more linear (as was generally the case with the categorical
 290 boundary theory) and it is predicted to have clear elbows at the prototype values.

291 Let's simulate the expected identification function for two categories with specific prototype values in one
 292 phonetic dimension. Let's say prototypes for $Cat_1 = 10$ and $Cat_2 = 20$, and the distance estimation function
 293 has the noise distribution ($\epsilon \sim N(\mu_\epsilon = 0, \sigma_\epsilon^2 = 5)$). We can now repeatedly simulate the categorization for a
 294 continuum of input values (say, 1000 times for each input value) and compare the observed identification curve

295 against the predicted identification curve from the above analytic result. And indeed for a low variance/noise
 296 ($\sigma_\theta^2 = 5$) in the distance function, the resulting identification function does “look” sigmoidal, though formally
 297 it is not (Figure 5).

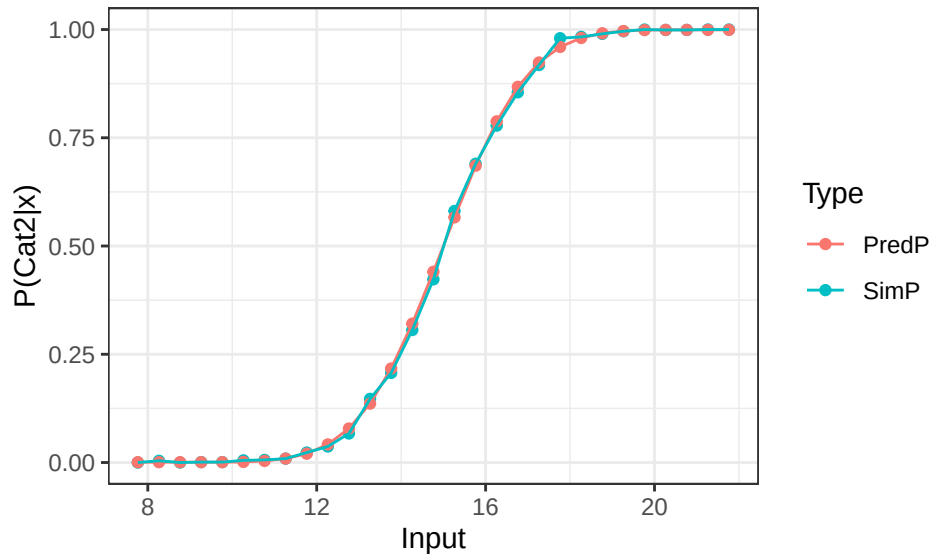


Figure 5: Comparing the $P(\text{Cat2}|\mathbf{x})$ from the analytic solution (salmon) against that from a simulation (cyan)

298 However, the problem becomes clear with increasing noise in the distance estimation function (Figure
 299 6). As the variance increases, the identification function moves further and further away from apparent
 300 sigmoidality to clearly showing elbows at the prototype points and steady-state values beyond the prototypes.
 301 Furthermore, the identification function becomes more linear in the range of the two prototype values.

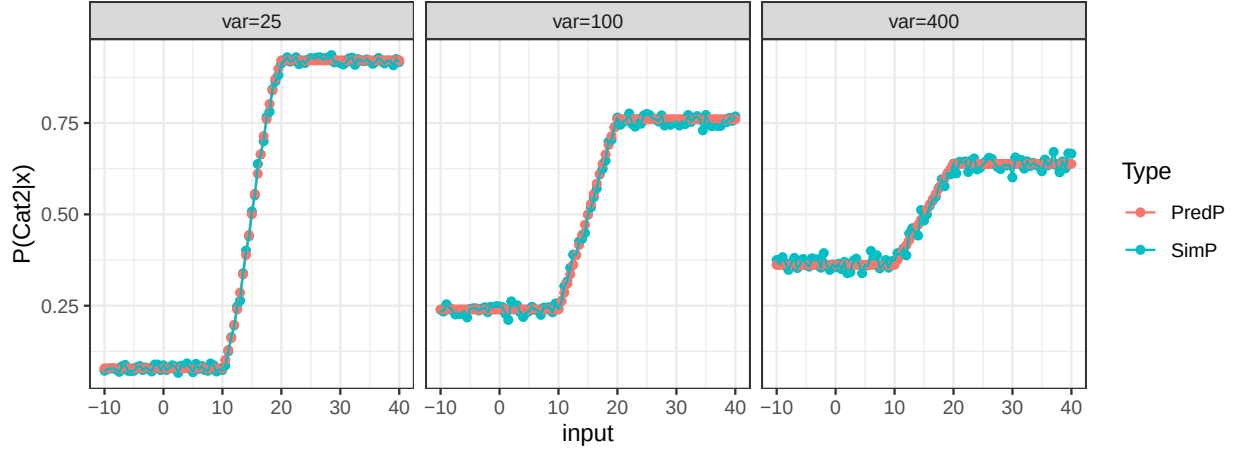


Figure 6: The identification function under Prototype theory with different amounts of distance measurement noise.

One could say that the steady-stating itself isn't an issue, and that the sharp transition in the probability function at the respective prototype points of the categories can be ameliorated by assuming that the input itself has a certain variance associated with it. Therefore, one could argue that this particular theory is also generally consistent with the phenomenological laws in (1). However, this theory of speech perception makes two rather surprising predictions: first, that the identification curve is steady-state beyond the means of the categories; second, that the minimum and maximum steady-state identification rates should be closer to 0.5 as the variance in the distance estimation increases.

Note, the analytic proof above was predicated on their being a single phonetic dimension. However, if the distance estimation noise is different but, crucially, independent in each of the k dimensions, then one can extend the same analytic expectation to each of the k dimensions. And crucially, there is no non-locality in the expected identification function in any of the k -dimensions.

2.2.1 What if prototype theory is cashed out in terms of similarities?

Another reasonable conception of prototype theories is in terms of *similarities* between the input and the prototypes, where similarities could be calculated as negative exponential functions of distances (Eq. 16) and the identification function is a relative proportional similarity calculation (Eq. 17) along the lines of Shepard (1957, 1958).

$$Sim(\mu, x) = exp^{-D(\mu, x)} \quad (16)$$

$$\mathbb{P}(Cat_2; x) = \frac{Sim(Cat_2, x)}{Sim(Cat_1, x) + Sim(Cat_2, x)} \quad (17)$$

318 It is easy to show that the thresholding and steady-state issue identified previously will remain. To
 319 make things clear, let's consider a no-noise case on a single phonetic dimensions, where Cat_2 has a higher
 320 prototype value than Cat_1 . In such a case, if the input x is higher in value than Cat_2 's prototype value,
 321 then the similarity values will be as follows:

$$\begin{aligned} Sim(Cat_1, x) &= \exp^{-D(\mu_1, x)} = \exp^{-|x - \mu_1|} = \exp^{-(x - \mu_1)} = \exp^{(\mu_1 - x)} \\ Sim(Cat_2, x) &= \exp^{-D(\mu_2, x)} = \exp^{-|x - \mu_2|} = \exp^{-(x - \mu_2)} = \exp^{(\mu_2 - x)} \end{aligned} \quad (18)$$

322 Note, in the above equations, since $x > \mu_1$ and $x > \mu_2$, $|x - \mu_1| = x - \mu_1$ and $|x - \mu_2| = x - \mu_2$.
 323 Consequently, the probability of Cat_2 is as follows:

$$\begin{aligned} \mathbb{P}(Cat_2; x) &= \frac{Sim(Cat_2, x)}{Sim(Cat_1, x) + Sim(Cat_2, x)} \\ &= \frac{\exp^{(\mu_2 - x)}}{\exp^{(\mu_1 - x)} + \exp^{(\mu_2 - x)}} \\ &= \frac{\exp^{-x} * \exp^{\mu_2}}{\exp^{-x} * \exp^{\mu_1} + \exp^{-x} * \exp^{\mu_2}} \\ &= \frac{\exp^{\mu_2}}{\exp^{\mu_1} + \exp^{\mu_2}} \end{aligned} \quad (19)$$

324 As can be seen in Eq. 19, when the input is higher in value than the Cat_2 prototype, then the expected
 325 probability of Cat_2 for a given input value is *independent* of the input value, and is a constant. A similar
 326 calculation will show that the probability of Cat_2 is a constant when the input value is below that of the
 327 Cat_1 prototype/mean value. Therefore, calculating probabilities based on similarities results in exactly the
 328 same threshold and steady-state identification function as the original difference based calculation.

329 While avoided in the interest of space, it can also be shown that when noise is added to the distance
 330 calculation, a similar effect of noise as in the distance-based calculation is achieved on the identification func-
 331 tion; namely, the slope of the identification function gets flatter with increased noise and on the steady-state
 332 probability values on either side of the category prototypes/means get closer with increased noise. Instead,
 333 in Figure 7, I simulate the identification functions for Prototype theories where similarity is calculated over

noisy distance measurements, and the Eq. 17 is used to get the probability of Cat_2 identification given an input x . The parameters are identical to those used in the simulation for Figure 6. As can be seen, the identification functions remain essentially identical to those in Figure 6. As can be seen, exactly the same pattern obtains as with the computation over distances earlier.

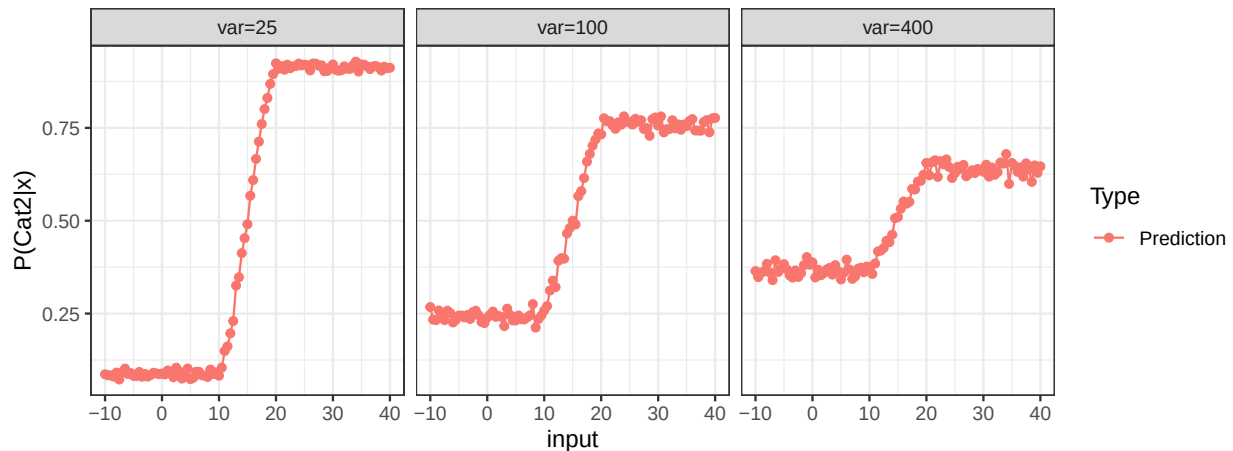


Figure 7: The identification function under Prototype theory using similarities with different amounts of distance measurement noise.

To summarize, the two issues, namely, steady-state identification functions beyond the mean/Prototype values of the categories along a phonetic dimension and the issue of decreasing distance between the maximum and minimum points along the identification curve with increasing noise, are both core characteristics of Prototype theories, whether Prototype theories are cashed out as differences over distances or as relative proportions over similarities.

2.3 Bayesian theories

In Bayesian categorization theories, Bayes theorem plays a crucial role in the computation. The initial formalization I present below largely follows that in Feldman and Griffiths (2007) and Feldman et al. (2009), with a few simplifications (when it doesn't affect the main point). The consequent analytic results are new, however.

For a 2-category system along a single phonetic dimension, we can model each of the categories as a random variable that is Normally distributed (20).⁷ Without loss of generality, we can assume that Cat_2

⁷Feldman et al. (2009) also include another random variable that represents the noise distribution $N(0, \sigma_{noise}^2)$ over a single

350 has the higher mean along the phonetic continuum.

$$\begin{aligned} Cat_1 &\sim N(\mu_1, \sigma_1^2) \\ Cat_2 &\sim N(\mu_2, \sigma_2^2) \end{aligned} \tag{20}$$

351 Given that each category has a Normal distribution, each of them has the following probability distribu-
352 tion function (p.d.f.) in (21).

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right) \tag{21}$$

353 Now, we can use Bayes theorem to calculate the probability of the identification of Cat_2 given a specific
354 input x as in Eq. 22, where $\mathbb{P}(Cat_i)$ refers to the prior probability of the category i .

$$\begin{aligned} \mathbb{P}(Cat_2|x) &= \frac{f(x|Cat_2) * \mathbb{P}(Cat_2)}{\sum_{i=1}^2 f(x|Cat_i) * \mathbb{P}(Cat_i)} \\ &= \frac{f(x|Cat_2) * \mathbb{P}(Cat_2)}{f(x|Cat_1) * \mathbb{P}(Cat_1) + f(x|Cat_2) * \mathbb{P}(Cat_2)} \\ &\text{dividing numerator and denominator by } f(x|Cat_2) * \mathbb{P}(Cat_2), \\ &= \frac{1}{1 + \frac{f(x|Cat_1) * \mathbb{P}(Cat_1)}{f(x|Cat_2) * \mathbb{P}(Cat_2)}} \\ &\text{replacing the likelihood with the corresponding p.d.f in (21),} \\ &= \frac{1}{1 + \frac{\frac{1}{\sigma_1\sqrt{2\pi}} e^{\left(-\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2\right)} * \mathbb{P}(Cat_1)}{\frac{1}{\sigma_2\sqrt{2\pi}} e^{\left(-\frac{1}{2}\left(\frac{x-\mu_2}{\sigma_2}\right)^2\right)} * \mathbb{P}(Cat_2)}} \\ &\text{simplifying,} \\ &= \frac{1}{1 + \frac{\sigma_2}{\sigma_1} * \frac{\mathbb{P}(Cat_1)}{\mathbb{P}(Cat_2)} * e^{\left(-\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2\right) - \left(-\frac{1}{2}\left(\frac{x-\mu_2}{\sigma_2}\right)^2\right)}} \end{aligned} \tag{22}$$

355 As can be seen from Eq. 22, the identification function depends on the prior probabilities, the means,
356 and the variances of the two categories. In standard presentations, Eq. 22 is reduced further by making a
target production. For our purposes, the distinction between *Target* and actual *exponence* is not relevant — the additional
random variable simply increases the variances of the categories, and can be subsumed in the variance that each category
already has. This simplification has no bearing on the results presented in this paper.

357 simplifying assumption that the variances of the two categories are equal ($\sigma_1 = \sigma_2 = \sigma$).

$$\begin{aligned}
 \mathbb{P}(Cat_2|x) &= \frac{1}{1 + \frac{\sigma}{\sigma} * \frac{\mathbb{P}(Cat_1)}{\mathbb{P}(Cat_2)} * e^{\left(-\frac{1}{2}\left(\frac{x-\mu_1}{\sigma}\right)^2\right) - \left(-\frac{1}{2}\left(\frac{x-\mu_2}{\sigma}\right)^2\right)}} \\
 &\text{expanding the square terms and re-organising,} \\
 &= \frac{1}{1 + \frac{\mathbb{P}(Cat_1)}{\mathbb{P}(Cat_2)} * e^{-\left(\frac{\mu_2-\mu_1}{\sigma}\right)*x + \left(\frac{\mu_2^2-\mu_1^2}{2\sigma}\right)}} \\
 &\text{since, } x = e^{\log(x)} \tag{23} \\
 &= \frac{1}{1 + e^{\log\left(\frac{\mathbb{P}(Cat_1)}{\mathbb{P}(Cat_2)}\right)} * e^{-\left(\frac{\mu_2-\mu_1}{\sigma}\right)*x + \left(\frac{\mu_2^2-\mu_1^2}{2\sigma}\right)}} \\
 &\text{(finally, since } e^x * e^y = e^{x+y}\text{)} \\
 &= \frac{1}{1 + e^{-\left(\frac{\mu_2-\mu_1}{\sigma}\right)*x + \left(\frac{\mu_2^2-\mu_1^2}{2\sigma}\right) + \log\left(\frac{\mathbb{P}(Cat_1)}{\mathbb{P}(Cat_2)}\right)}}
 \end{aligned}$$

358 In Eq. 23, except for the input value x , everything else is a constant. Therefore, the equation has the
 359 form of a logistic function in (24), and the logistic function guarantees a sigmoidal curve that asymptotes
 360 to 0 and 1 on either side. Notably, the monotonicity and sigmoidality are guaranteed no matter what the
 361 means, variances and priors are, as long as the variances are identical for the two categories. Furthermore,
 362 there will be a unique category boundary that represents $\mathbb{P}(Cat_2|x) = \mathbb{P}(Cat_1|x) = 0.5$.

$$\mathbb{P}(Cat_2|x) = \frac{1}{1 + \exp^{-A*x+B}} \tag{24}$$

363 This standard presentation is often used to argue that Bayesian theories correctly predict a sigmoidal
 364 curve. However, the proof crucially depends on the assumption of equal variances of categories. Notably,
 365 the assumption of equal variance of categories runs contrary to what researchers in the Bayesian mould
 366 have argued for in attempts to account for differences between the identification functions of vowels versus
 367 consonants (Feldman and Griffiths 2007; Feldman et al. 2009; Kronrod et al. 2016; Yu 2011), and in attempts
 368 to account for the phenomena of perceptual returning and selective adaptation (Kleinschmidt and Jaeger
 369 2015; Tan and Jaeger 2025). So the issue of the explanatory value of Bayesian theories in regards to the
 370 phenomenological laws in (1) needs to be studied in the broader context of unequal variances of categories. As

will be seen below, the assumption of equal variances of categories is neither trivial nor innocent. Particularly, it will be shown that when the variances are not equal, the identification curve will be either convex downward or concave downward, and not monotonic or sigmoidal.

2.3.1 Where is the categorical boundary in the case of Bayesian theories?

In order to show the point alluded to at the end of the previous sub-section, we will start at a different point of departure than the standard proof shown in the previous sub-section. Using the Eq. 23, we can define both the relevant conditional probabilities.

$$\begin{aligned}\mathbb{P}(Cat_1|x) &= \frac{1}{1 + \frac{\sigma_1}{\sigma_2} * \frac{\mathbb{P}(Cat_2)}{\mathbb{P}(Cat_1)} * e^{\left(-\frac{1}{2}\left(\frac{x-\mu_2}{\sigma_2}\right)^2\right) - \left(-\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2\right)}} \\ \mathbb{P}(Cat_2|x) &= \frac{1}{1 + \frac{\sigma_2}{\sigma_1} * \frac{\mathbb{P}(Cat_1)}{\mathbb{P}(Cat_2)} * e^{\left(-\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2\right) - \left(-\frac{1}{2}\left(\frac{x-\mu_2}{\sigma_2}\right)^2\right)}}\end{aligned}\quad (25)$$

We can now define the categorical boundary as the point where $\mathbb{P}(Cat_2|x) = \mathbb{P}(Cat_1|x)$. Therefore, from (25),

$$\frac{1}{1 + \frac{\sigma_2}{\sigma_1} * \frac{\mathbb{P}(Cat_1)}{\mathbb{P}(Cat_2)} * e^{\left(-\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2\right) - \left(-\frac{1}{2}\left(\frac{x-\mu_2}{\sigma_2}\right)^2\right)}} = \frac{1}{1 + \frac{\sigma_1}{\sigma_2} * \frac{\mathbb{P}(Cat_2)}{\mathbb{P}(Cat_1)} * e^{\left(-\frac{1}{2}\left(\frac{x-\mu_2}{\sigma_2}\right)^2\right) - \left(-\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2\right)}} \quad (26)$$

Eq. 26 above can be reduced and then used to solve for x , namely the categorical boundary. This results in two cases (See A.1 for details on the reduction).

Case 1: if $(\sigma_1^2 - \sigma_2^2) = 0$

$$x = \frac{\sigma_1^2 \sigma_2^2 \log\left(\frac{\sigma_1}{\sigma_2} \frac{\mathbb{P}(Cat_2)}{\mathbb{P}(Cat_1)}\right)^2 + \sigma_2^2 \mu_1^2 - \sigma_1^2 \mu_2^2}{2 * (\sigma_2^2 \mu_1 - \sigma_1^2 \mu_2)} \quad (27)$$

Case 2: if $(\sigma_1^2 - \sigma_2^2) \neq 0$

$$x = \left(\frac{\sigma_1^2 \mu_2 - \sigma_2^2 \mu_1}{\sigma_1^2 - \sigma_2^2}\right) \pm \sqrt{\frac{\sigma_1^2 \sigma_2^2 \log\left(\frac{\sigma_1}{\sigma_2} \frac{\mathbb{P}(Cat_2)}{\mathbb{P}(Cat_1)}\right)^2 + \sigma_2^2 \mu_1^2 - \sigma_1^2 \mu_2^2}{(\sigma_1^2 - \sigma_2^2)} + \left(\frac{\sigma_2^2 \mu_1 - \sigma_1^2 \mu_2}{\sigma_1^2 - \sigma_2^2}\right)^2} \quad (28)$$

Case 1 in Eq. 27 suggests, consistent with standard presentations, that when the variances of the two categories are equal, there is indeed a single categorical boundary. However, Case 2 in Eq. 28 suggests that when the variances of the two categories are not equal, then there are *two* categorical boundaries; *i.e.*, two points in the identification curve where the $Prob(Cat_1|x) = Prob(Cat_2|x) = 0.5$. Furthermore, given the fact that there are two distinct categorical boundary points, the identification curve cannot be monotonic or sigmoidal.

2.3.2 First simulation of Bayesian theories

In order to gain some intuition about the results, it is helpful to simulate identification curves for each of the two cases in Eqs. 27-28. Table 1 lists the different sets of simulation parameters used.

Table 1: Different sets of simulation parameters.

mean1	mean2	var1	var2	P1	P2
10	40	100	100	0.5	0.5
10	40	100	400	0.5	0.5

The predicted identification functions are shown for each set in Figure 8. For the parameter set where the variances of the two categories are equal (Simulation 1), the simulation generates an identification function that is monotonically increasing, and is sigmoidal curve. However, as expected per the proof above, when the variances of the two categories are not identical, even the weak law of monotonicity is violated (Simulation 2), and in fact there are *two* categorical boundaries (50% points) in such cases.

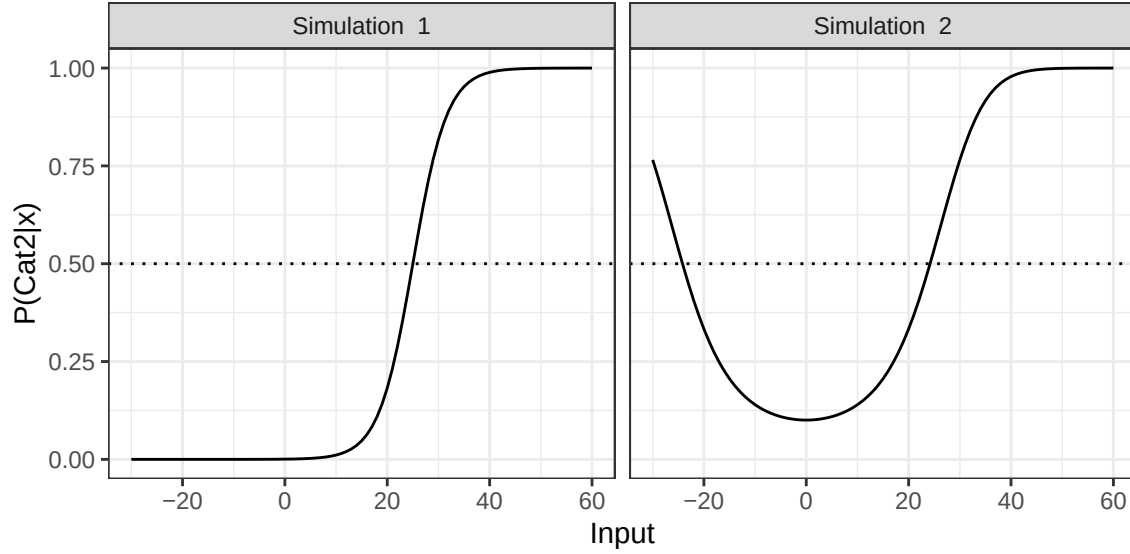


Figure 8: Predicted identification functions for Bayesian theories. Simulation 1 = equal variances; Simulation 2: Unequal variances. The dotted horizontal line marks the categorical boundary.

2.3.3 What are the inflection points in the identification curve?

In the previous sub-section, we saw that when the variances are unequal, the identification function *must* have two categorical boundaries. Furthermore, we saw that in such a case, there is an “inflection point” between the two categorical boundaries, where the identification curve changes directions. We can now push the formalism further to identify the *inflection point*, defined as the point where there is a “bend” in the identification curve.

We can return to Eq. 22 which is re-organized below.

$$\begin{aligned}
 \mathbb{P}(Cat_2|x) &= \frac{1}{1 + \frac{\sigma_2}{\sigma_1} * \frac{\mathbb{P}(Cat_1)}{\mathbb{P}(Cat_2)} * e^{\left(-\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2\right) - \left(-\frac{1}{2}\left(\frac{x-\mu_2}{\sigma_2}\right)^2\right)}} \\
 &= \frac{1}{1 + e^{\left(-\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2\right) + \left(\frac{1}{2}\left(\frac{x-\mu_2}{\sigma_2}\right)^2\right) + \log\left(\frac{\sigma_2}{\sigma_1} * \frac{\mathbb{P}(Cat_1)}{\mathbb{P}(Cat_2)}\right)}}
 \end{aligned} \tag{29}$$

And now, we can get the first derivative of $\mathbb{P}(Cat_2|x)$, equate it to 0, and solve for x to identify the

inflection point.

$$\frac{d}{dx}(\mathbb{P}(Cat_2|x)) = \frac{d}{dx} \frac{1}{1 + e^{\left(-\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2\right) + \left(\frac{1}{2}\left(\frac{x-\mu_2}{\sigma_2}\right)^2\right) + \log\left(\frac{\sigma_2}{\sigma_1} * \frac{\mathbb{P}(Cat_1)}{\mathbb{P}(Cat_2)}\right)}} = 0 \quad (30)$$

Simplifying and solving for x (see Appendix A.2 for the full derivation), we get the result in Eq. 31 under the assumption that both variances are non-zero and different.

$$x = \frac{\sigma_1^2 * \mu_2 - \sigma_2^2 * \mu_1}{\sigma_1^2 - \sigma_2^2} \quad (31)$$

Eq. 31 gives us the inflection point when the two category variances are different (and non-zero). This result guarantees that whenever $\sigma_1 \neq \sigma_2$, there *must be* a minimal or maximal inflection point along the identification function. Furthermore, when $\sigma_1 = \sigma_2$, there is no inflection point, and so we are able to retain the sigmoidal curve. A particularly important observation here is that the inflection point is independent of the prior probabilities of the categories, and depends *only* on the variances and means of the categories. With this result, we have an understanding of why and where there is an inflection point in the identification function. If Cat_2 has a higher variance than Cat_1 , then even if Cat_2 has a higher mean than Cat_1 , there will still be two points beyond which the densities for Cat_2 will necessarily be higher than for Cat_1 . That is, the presence of the inflection point and the existence of two category boundary values is due to differences in the likelihood functions themselves (and are is not dependent on the priors of the categories). Consequently, any form of distance comparison between the input and a pair of categories involving likelihood functions will have the same issue.

In this sub-section, while we have established that there is necessarily a single inflection point when the two variance are unequal, we have not obtained a formal proof for the shape of the identification curve when the variances are unequal.

2.3.4 Shape of the identification curve when the variances are unequal: convex vs. concave

We can probe the convexity/concavity of the identification function, by looking at the sign of the second derivative at the inflection point ($x = \frac{\sigma_1^2 * \mu_2 - \sigma_2^2 * \mu_1}{\sigma_1^2 - \sigma_2^2}$). If the second derivative at that point is positive, then

the inflection point reflects a minima, and therefore, the shape of the identification curve must be convex downwards (Tanner 2019). Alternatively, if the second derivative at the inflection point is negative, then the inflection point is a maxima, and therefore, the shape of the identification curve is concave downwards.

Let's get the second derivative of the categorization function.

$$\begin{aligned}
 \frac{d^2}{dx^2} \mathbb{P}(Cat_2|x) &= \frac{d}{dx} \frac{e^{\left(-\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2\right) + \left(\frac{1}{2}\left(\frac{x-\mu_2}{\sigma_2}\right)^2\right) + \log\left(\frac{\sigma_2}{\sigma_1} * \frac{\mathbb{P}(Cat_1)}{\mathbb{P}(Cat_2)}\right)} * \left(\left(\frac{x-\mu_1}{\sigma_1}\right) - \left(\frac{x-\mu_2}{\sigma_2}\right)\right)}{\left(1 + e^{\left(-\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2\right) + \left(\frac{1}{2}\left(\frac{x-\mu_2}{\sigma_2}\right)^2\right) + \log\left(\frac{\sigma_2}{\sigma_1} * \frac{\mathbb{P}(Cat_1)}{\mathbb{P}(Cat_2)}\right)}\right)^2} \\
 &= \frac{\left(\frac{1}{\sigma_1^2} - \frac{1}{\sigma_2^2}\right)}{\left(2 + e^{\left(\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2\right) - \left(\frac{1}{2}\left(\frac{x-\mu_2}{\sigma_2}\right)^2\right) - \log\left(\frac{\sigma_2}{\sigma_1} * \frac{\mathbb{P}(Cat_1)}{\mathbb{P}(Cat_2)}\right)} + e^{\left(-\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2\right) + \left(\frac{1}{2}\left(\frac{x-\mu_2}{\sigma_2}\right)^2\right) + \log\left(\frac{\sigma_2}{\sigma_1} * \frac{\mathbb{P}(Cat_1)}{\mathbb{P}(Cat_2)}\right)}\right)} \\
 &\quad + \left(\left(\frac{x-\mu_1}{\sigma_1^2}\right) - \left(\frac{x-\mu_2}{\sigma_2^2}\right)\right)^2 * \\
 &\quad \left(\frac{e^{\left(-\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2\right) + \left(\frac{1}{2}\left(\frac{x-\mu_2}{\sigma_2}\right)^2\right) + \log\left(\frac{\sigma_2}{\sigma_1} * \frac{\mathbb{P}(Cat_1)}{\mathbb{P}(Cat_2)}\right)} - e^{\left(\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2\right) - \left(\frac{1}{2}\left(\frac{x-\mu_2}{\sigma_2}\right)^2\right) - \log\left(\frac{\sigma_2}{\sigma_1} * \frac{\mathbb{P}(Cat_1)}{\mathbb{P}(Cat_2)}\right)}}{\left(2 + e^{\left(\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2\right) - \left(\frac{1}{2}\left(\frac{x-\mu_2}{\sigma_2}\right)^2\right) - \log\left(\frac{\sigma_2}{\sigma_1} * \frac{\mathbb{P}(Cat_1)}{\mathbb{P}(Cat_2)}\right)} + e^{\left(-\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2\right) + \left(\frac{1}{2}\left(\frac{x-\mu_2}{\sigma_2}\right)^2\right) + \log\left(\frac{\sigma_2}{\sigma_1} * \frac{\mathbb{P}(Cat_1)}{\mathbb{P}(Cat_2)}\right)}\right)^2} \right)
 \end{aligned} \tag{32}$$

The second derivative in Eq. 32 happens to be a rather ugly and unwieldy function. However, when we plug in the value of x at the inflection point ($x = \frac{\sigma_1^2 * \mu_2 - \sigma_2^2 * \mu_1}{\sigma_1^2 - \sigma_2^2}$), the equation simplifies dramatically. See Appendix A.3 for full derivation.

$$\begin{aligned}
 \frac{d^2}{dx^2} \mathbb{P}\left(Cat_2|x = \frac{\sigma_1^2 * \mu_2 - \sigma_2^2 * \mu_1}{\sigma_1^2 - \sigma_2^2}\right) &= \frac{\left(\frac{1}{\sigma_1^2} - \frac{1}{\sigma_2^2}\right)}{\left(2 + e^{\left(\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2\right) - \left(\frac{1}{2}\left(\frac{x-\mu_2}{\sigma_2}\right)^2\right) - \log\left(\frac{\sigma_2}{\sigma_1} * \frac{\mathbb{P}(Cat_1)}{\mathbb{P}(Cat_2)}\right)} + e^{\left(-\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2\right) + \left(\frac{1}{2}\left(\frac{x-\mu_2}{\sigma_2}\right)^2\right) + \log\left(\frac{\sigma_2}{\sigma_1} * \frac{\mathbb{P}(Cat_1)}{\mathbb{P}(Cat_2)}\right)}\right)} \\
 &\tag{33}
 \end{aligned}$$

The crucial observation is that the denominator in Eq. 33 is always positive, so the sign of the second derivative at the inflection point will depend on the numerator. The numerator can be transformed to a more comprehensible form.

$$\text{numerator of Eq. 33} = \left(\frac{1}{\sigma_1^2} - \frac{1}{\sigma_2^2}\right) = \frac{\sigma_2^2 - \sigma_1^2}{\sigma_1^2 * \sigma_2^2} \tag{34}$$

From the transformed numerator in Eq. 34, we can see that when $\sigma_1^2 > \sigma_2^2$, the numerator is positive. This means that the second derivative at the inflection point is positive; therefore, the identification function is convex downwards. However, when $\sigma_1^2 < \sigma_2^2$, the second derivative at the inflection point is negative; therefore, the identification function is concave downwards. This reiterates that when the variances are unequal, the identification function is neither sigmoidal nor monotonic. Finally, when $\sigma_1^2 = \sigma_2^2$, the second derivative at the inflection point is 0, therefore the categorization function is neither convex nor concave — this is consistent with the fact that when the variances are equal, the identification curve is sigmoidal.

2.3.5 Second simulation of Bayesian theories

Now, we are in a position to simulate a variety of parameter values to get more intuitive understanding of the results. The current simulations add to those in Section 2.3.2. Table 2 lists the different sets of simulation parameters used.

Table 2: Different sets of simulation parameters and the corresponding inflection point for Bayesian categorization.

Set	mean1	mean2	var1	var2	P1	P2	inflectionPoint
Simulation 1	10	40	100	100	0.50	0.50	Inf
Simulation 2	10	40	100	400	0.50	0.50	0
Simulation 3	10	40	100	100	0.75	0.25	Inf
Simulation 4	10	40	100	400	0.25	0.75	0
Simulation 5	10	40	100	400	0.75	0.25	0
Simulation 6	10	40	400	100	0.50	0.50	50

The predicted identification functions are shown for each set in Figure 9. The first two simulations are repeated from Section 2.3.2, and show that the the identification function is neither sigmoidal nor monotonic when the variances are unequal. In fact, it is convex or concave depending on which variance is larger, and there are two categorical boundaries (50% points) for the identification function. This observation is present in Simulations 4-6. Simulations 1 and 3 show that both monotonicity and sigmoidality are expected

for different priors when the variances are identical. Simulations 2, 4 and 5 show that the inflection point is not affected by the priors of the categories when the variances are not identical. Next, Simulation 6 in comparison to Simulations 2, 4, 5 shows that the identification function is convex downwards when the second category (defined as the one with the higher mean) has more variance, and is concave downwards when the first category has more variances. Finally, the simulations also shows that minimum/maximum probability of identification of categories depends on the prior probabilities of the categories and the variances of the categories — if the prior probability of a category decreases, then the minimum identification probability for that decreases; similarly, the more the discrepancy in the variances, the more the minimum/maximum identification probabilities (as the case maybe) is away from the 0/1 values.

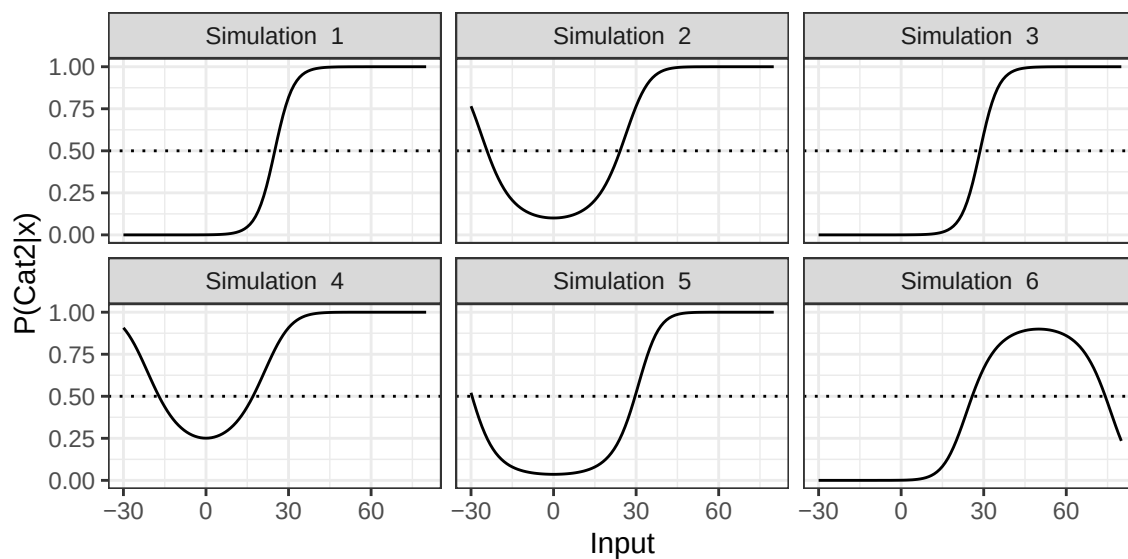


Figure 9: Predicted identification functions for different sets of parameter. The dotted horizontal line marks the categorical boundary.

The simulations fully replicate all the crucial analytic results in the previous sub-sections. Furthermore, it is quite clear at this point that when the variances of the two categories are unequal, the identification function is neither sigmoidal nor monotonic. Furthermore, the identification shapes are further complicated by differences in the prior probabilities of the categories.

2.4 Exemplar theories

Exemplar theories of perception depend on the relative average distance of the input signal from the cluster of stored exemplars for a category (Goldinger 1996, 1998; Nosofsky 1987, 1988a,b). In one instantiation of this idea for a 2-category system, Nosofsky (1988b) suggests the following equation to get the probability of a category given a specific input (x), where $bias_i$ is a free parameter, $0 \leq bias_i \leq 1$, that represents the *a priori* preference for a category (akin to a prior in a Bayesian theories).

$$\mathbb{P}(Cat_1|x) = \frac{bias_1 * \sum_{e \in C_1} \eta_{1x}}{bias_1 * \sum_{e \in C_1} \eta_{1x} + (1 - bias_1) * \sum_{e \in C_2} \eta_{2x}} \quad (35)$$

In Eq. 35, e_i refers to each exemplar of Cat_i , and η_{ix} refers to the similarity of a single exemplar from Cat_i to the input. Based on prior work studying the relationship between distance and similarity (Nosofsky 1987; Shepard 1957, 1958, 1986), Nosofsky (1988b) suggested that similarity be calculated as the scaled negative exponent of the distance between the exemplar of that category along all the relevant phonetic dimensions and the input, where the scaling factor c is a free parameter such that $0 \leq c \leq \infty$.

$$\eta_{ix} = exp^{-cd_{ix}} \quad (36)$$

Note, the distance measure itself is calculated as the Euclidean distance between each exemplar (e_{ik}) and the input in $k = 2$ phonetic dimensions, where each phonetic dimension has an associated attentional weight (w_k).

$$d_{ix} = \sqrt{w_1(e_{i1} - x_1)^2 + (1 - w_1)(e_{i2} - x_2)^2} \quad (37)$$

Finally, there is also a parameter associated with the frequencies of each of the exemplars, which I have avoided in the interest of expository and analytic ease. In what follows, I will assume that the exemplars are equally frequent in the interest of concision — this assumption, if anything, reduces the space of possible identification contours, and therefore doesn't disadvantage the theory in any way with respect to formal comparison with other theories. As can readily be seen from the above set of equations, the actual perceptual predictions of exemplar theories depend of the exact *distribution* of the exemplars under each category.

However, the distributions of exemplars have no specific constraint on them, beyond perhaps the limits imposed by human anatomy and neural mechanisms. Consequently, it is difficult to formally analyse exemplar theories and obtain analytic results which are not self-evident from the relevant formulas above. Furthermore, researchers have argued that this flexibility is what gives exemplar representations the expressive power to describe observed patterns in the production of individual lexical items (Goldrick and Cole 2023; Hay et al. 2017; Pierrehumbert 2002, 2016). Therefore, it is the very nature of exemplar theories that prevents the attainment of formal/analytic results — minimally, it is beyond the technical abilities of the current author to derive such analytic results.

For this reason, in order to understand the possible shapes of identification functions that are possible under exemplar theories, I simulate some potential input distributions, to see if Exemplar theories of speech categorization can predict or explain either the weak law or the strong law of speech categorization. As a result, I present existence proofs of the violation of the monotonicity and sigmoidality in the expected identification functions under a variety of input distributions.

To simplify the simulations, I will present the results for a continuum of input values for a two category system in a single phonetic dimension. It's assumed that the bias parameters ($bias_i$) for each category are identical. Furthermore, the category distributions are modelled as mixtures of equiprobable normal distributions (for the purposes of the replicability of the observed patterns). Each category was simulated with 3000 equally frequent exemplars. In cases where a category distribution is modelled as the mixture of multiple distributions, each of the constituent distributions had an equal number of exemplars such that the total amounts to 3000.

Simulation #	Category 1	Category 2
1	$N(-5, 1)$	$N(0, 1)$
2	$N(-5, 1)$	$N(0, 100)$
3	$\{N(-5, 1), N(5, 1)\}$	$N(0, 1)$
4	$\{N(-5, 1), N(5, 1), N(15, 1)\}$	$\{N(0, 1), N(10, 1)\}$
5	$\{N(-5, 1), N(5, 1)\}$	$N(0, 25)$

Table 3: The means and variances used for each category for each set of simulations for the Exemplar theory of perception.

As can be seen from the simulation results in 10, when the input distribution already has the property of being derived from simple Normally distributed categories (first facet in the figure), then we observe monotonicity and sigmoidality. However, the identification function is not generally monotonically increasing/decreasing or sigmoidal, as is observed in the other cases. In fact, there are some scenarios where the identification probability never falls below 0.5. Furthermore, in the second facet of the figure, where Cat_2 has a wider distribution than Cat_1 , the keen observer will notice that the identification function rises slightly as we go in the negative direction from 0 along the input continuum. Furthermore, unlike the Bayesian results in Section 2.3, there is no guarantee of even a single inflection point in the identification function. In fact, multiple inflection points are possible based on the distributions of the categories.

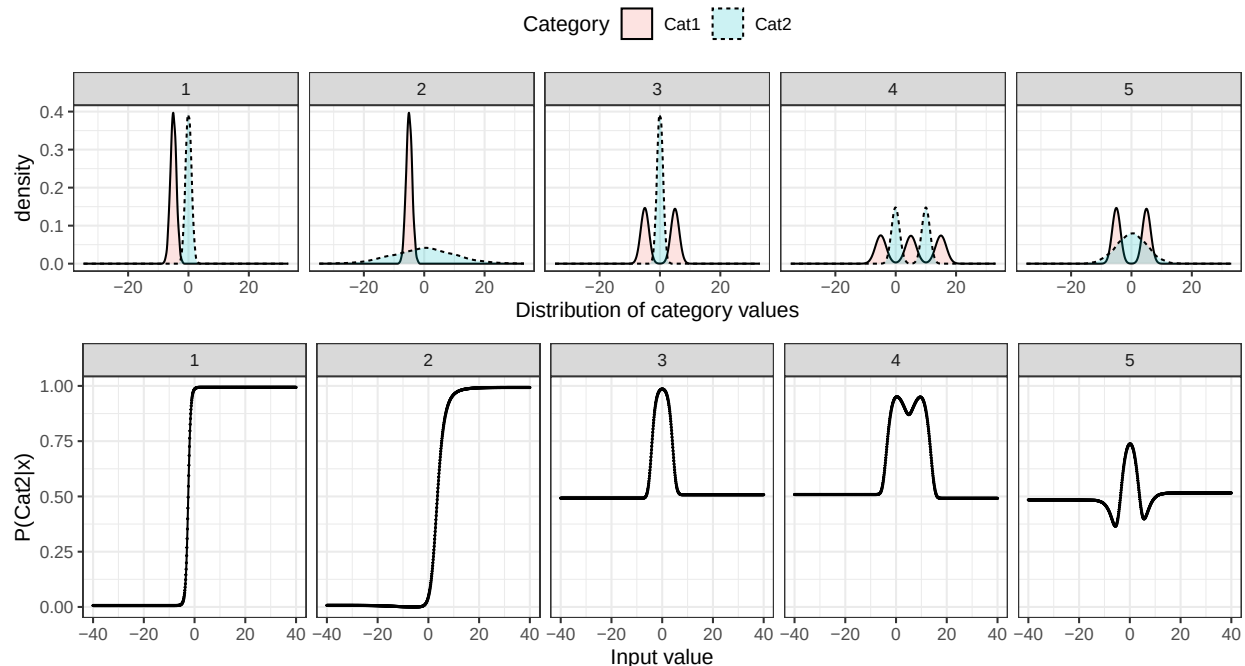


Figure 10: The identification functions for different distributions of Exemplar categories

The only situation in which Exemplar theories predict the weak and strong laws of speech categorization are cases in which the category distributions are well-behaved normal distributions, in which case the category distributions can be reduced to just two parameters, *i.e.*, exemplar theories of perception correctly capture the weak and strong laws of speech categorization *only* when the categories are (roughly) normally distributed. In essence, the right predictions follow *only* when category distributions are modelled as stemming from abstract categories that are related to a unique set of parameters related to Normal distributions.

Furthermore, it is to be noted that in the above simulations, the category exemplar distributions used are still quite well-behaved, as they have been generated from mixtures of Normal distributions. Therefore, when we allow ourselves the full expressive power of Exemplar theories by removing non-justified restrictions on the category distributions, we can see that the space of possible shapes of the identification is impossibly large, and therefore, the theory neither predicts nor explains either the weak law or the strong law of speech categorization. Furthermore, the theory has a lot of free parameters that make it that enlarge the space of possibilities, and allow for curve-fitting.

3 Discussion and Conclusion

The focus of the current paper was to formally compare four theories well represented in the literature in regards to how well they explain the weak and strong laws of speech categorization, restated below in (3).

3. Two laws of speech categorization

(a) **Weak law:** The categorization function increases/decreases monotonically increasing along a phonetic dimension.

(b) **Strong law:** The categorization function has a sigmoidal shape along a phonetic dimension.

The results for this primary question are summarized in Table 4.⁸ The only theories of speech categorization to actually explain both the laws were the Category Boundary theories. Such theories truly explain the laws because the theories *entail* them, and cannot accommodate any observations to the contrary. While Prototype theories were able to explain the weak law, they did not successfully explain the strong law. Finally, both Bayesian and Exemplar theories fail on both counts. In fact, both of the latter theories organically predict “non-local” categories where the ranges of the categories are disjoint. Such non-locality is never observed naturally for speech categories, and it has been argued recently that such non-local categories are substantially more difficult for humans to learn in artificial language paradigms (Heffner et al. 2019), suggesting that learners might use other additional puzzle-solving strategies in such tasks that are not involved in natural speech category learning. A possible extension of the current results not shown here is that the resulting categorical boundary necessarily has to be a hyperplane in k-dimensions to guarantee monotonicity. If there is any curvature in the categorical boundary, then non-monotonicity of the identification function (and consequently, non-locality of categories) will be guaranteed. Consequently, only theories that guarantee that the categorical boundaries will be hyperplanes, namely, Category Boundary theories and Prototype theories, have the mathematical structure to explain this observation about naturally occurring speech categories.

⁸In the Appendix A.4, I use real production data from an experiment studying Voice Onset Times (VOTs) in American English to estimate perception parameters for the four theories and then test whether they correctly obtain the expected shape of the identification function.

Theory	Weak law	Strong law
Category Boundary theories	✓	✓
Prototype theories	✓	✗
Bayesian theories	✗	✗
Exemplar theories	✗	✗

Table 4: Do the theories explain the weak and strong laws of speech categorization?

The success of simple Categorical Boundary theories is particularly relevant for any theory of perception that espouses discrete representations. The formal modelling here suggests that the currently best way to understand the sigmoidal average response curves in identification tasks is in terms of identifying discrete categories. See Caplan and Durvasula ([under review](#)) for a recent algorithmic implementation of the idea. While I’ve focussed narrowly on speech categorization throughout the paper, it is possible to extend a similar view of discrete categories to other domains where sigmoidal curves have been observed, such as color categorization (Beare 1963; Bornstein and Korda 1985; Bornstein 1987; Huettenlocher 2012) and to historical language change (Kroch 1989; Labov 2001), language acquisition (Yang 2001), and phonotactics (Hayes 2022), amongst other fields

In addition to establishing that primary point, the paper also revealed many new properties for the theories of speech categorization considered. In laying out these additional predictions, I hope to engender a constructive dialogue in future research. As with all models/theories, any mismatch between the relevant phenomenon and theoretical predictions need not be a death knell for the theory (Lakatos 1970). One can always rescue the theory by altering specific auxiliary hypotheses. Therefore, the challenge for Prototype, Bayesian and Exemplar theories is to precisely state what the modifications are so that one may assess them formally. All one can say is that in their current forms, they are not explanatory. Furthermore, in the formalization, I used very basic elements of the theories to get to the results of the expected space of identification function, yet the theories were found wanting. While the import and interpretation of such a fact will ultimately depend on the researcher, I interpret it as suggesting that the three theories have the wrong formal structures to model speech categorization. Having said that, in what follows I discuss some

possible ways to proceed for such theories.

For Category Boundary theories, the explanatory success with the two laws is a good sign. The obvious next challenges are in accounting for more complex phenomena: Perceptual learning (Nelson and Durvasula 2021; Norris et al. 2003; Vroomen et al. 2004), Ganong effect (Ganong 1980; Rysling et al. 2015), McGurk Effect (Dong et al. 2025; McGurk and MacDonald 1976; Van Wassenhove et al. 2007). With respect to perceptual learning, one could model this as a change in the mean value or the variance related to a specific categorical boundary; however, given categorical boundaries are the arbiters of categorization in the theory, this way of modelling learning will have a consequence on how perceptual learning affects other categories. How one extends the formalism to account for integration with other information, Ganong effect or McGurk Effect, will have to be the focus of future inquiry.

For Prototype theories, the formal analysis and simulation revealed that the theory makes two rather surprising predictions: first, that the identification function steady states beyond the mean (prototype) category values; second, that the steady state values get closer to each other and 0.5 (pure random guess) as the variances in distance measurement increases. It should be noted that if the distance noise/variance is sufficiently small, then the identification function might look superficially similar to a sigmoidal function, as was shown in the paper; however, formally, it is never a sigmoidal function. Therefore, this set of predictions sets up the possibility of future testing for researchers who are sympathetic to Prototype theory. One could therefore reject the sigmoidal characterization of the identification function and show that the shape and range of the identification function varies with additional noise/variance in the distance function. As a note of skepticism of this possibility, it is worth pointing out that despite there being more noise in vowel distributions in production, it has been observed since the earliest work on vowel categorization (Fry et al. 1962; Pisoni 1973; Pisoni and Lazarus 1974) that the vowel identification functions often reach either 0% or 100% identification rates at the relevant ends of the phonetic continuum. However, one may object that vowels and consonants do not differ in measurement noise; if so, it is worth thinking about other ways to change measurement noise so as to test the predictions of Prototype theories. Relatedly, in recent work, Stern (2021) observed that listeners' goodness judgements for some vowel category exemplars increase substantially past the prototype or mean value for the category. To the extent that goodness values are related to the

category identification (Lotto et al. 1998), the result is a problem for Prototype theories.

For Bayesian theories, the formal analysis and simulations showed that the sigmoidal curve was predicted, only under the condition of equal variances of categories. However, any time the categories have unequal variances, the identification function *will be* non-monotonic and non-sigmoidal, *i.e.*, any time the categories have unequal variances, one of the categories *will be* paired with a non-local set of values along a phonetic continuum. Next, it was also proved that the “inflection point” of the identification curve is not sensitive to category priors, suggesting that the problem of non-monotonicity stems from the likelihood functions associated with the categories, themselves. That is, the result suggests that the ultimate problem is with the use of likelihood functions in order to assess the relative distance between a specific input and a set of categories. Third, it was proved that the shape of the identification function (convex or concave) depends on which of the categories has the higher variance. Finally, it was proved that the maxima and minima of the identification curves are affected by the variances of the categories and by the prior probability. These predictions are quite precise and will benefit from future testing.

For now, it is worth reiterating that Bayesian theories are currently not explanatory, when it comes to the two laws restated in (3). One might attempt to salvage the theory by claiming that category variances are indeed identical in a 2-category identification task. However, then *ipso facto* all relevant categories in a language will be predicted to have the same category variances — which is a rather strong assumption in need of empirical demonstration. Furthermore, as noted above, the claim runs contrary to what researchers in the Bayesian mould have argued for in attempts to account for differences between the identification functions of vowels versus consonants (Feldman and Griffiths 2007; Feldman et al. 2009; Kronrod et al. 2016; Yu 2011), and in attempts to account for the phenomena of perceptual returning and selective adaptation (Kleinschmidt and Jaeger 2015; Tan and Jaeger 2025). Similarly, one needs to constrain the effects of priors so as to not allow for the undesirable possibilities, and again category priors have been used to account for the phenomena of perceptual returning and selective adaptation (Kleinschmidt and Jaeger 2015; Tan and Jaeger 2025). So, these phenomena will need alternative accounts. As they stand, they cannot be considered explanatory accounts of the relevant phenomena. Returning to the possible restriction of category variances being equal; if true, this is a rather interesting and strong restriction on category distributions that is

entirely worth exploring. However, doing so along with equating the priors effectively formally collapses the distinction between Bayesian theories and Category Boundary theories, and therefore is not an innocent theoretical choice.

A second possible approach is to threshold the category likelihoods/distributions to have finite ranges. Note, however, that the crucial problem identified in the formal analysis is that the likelihood of the category with the larger variance will be higher than that of the other category beyond a certain range. Therefore, to implement the idea of finite thresholds on distributions, one will also have to guarantee that the likelihood of one category is always lower than that of another category beyond a point. At this point, it is not clear how one effects that guarantee across the whole set of categories in a language.

A third possible approach is to try another variance distribution. For example, some have argued that Log-Normal distributions are perhaps better fits to many natural distributions often claimed to be Normal originally (Limpert et al. 2001). However, if the likelihood of one of two categories is higher for some input on the Normal scale, the likelihood must also be higher for the same category on the Log-Normal scale. And, therefore, the issue pointed out for Bayesian theories will persist with Log-Normal distributions. The real issue as far as I can see for Bayesian theories is that the variances are attached to the categories, unlike with Category Boundary theory and Prototype theory.

A final possibility is to limit the categorical boundary to be between the means of the categories. But, apart from stating it as such, it is unclear what principled basis we can use to constrain the categorical boundaries in this fashion. So, it would be (currently) an *ad hoc* restriction on the theory.

For Exemplar theories, the problems identified in the current paper are the worst, in a sense. The fundamental issue is that Exemplar theories are beholden to the actual category distributions. While in the past, the possibility of arbitrariness of category distributions was seen as a virtue of exemplar theories (Goldrick and Cole 2023; Hay et al. 2017; Pierrehumbert 2002, 2016), the virtue is only in *accounting for* observed patterns, and not explaining them. If our interest is in the latter, then there need to be severe restrictions on such category distributions. One possibility raised in the paper is that category be constrained to be Normally distributed. However, the question that arises is: how can one guarantee this? It will not do to nod to the fact that Normal distributions naturally arise in nature (Lyon 2014). The basis of Normality

in nature is often thought to be that the observed distribution stems from the additive effect of many sub-distributions, all of which have variances that are much smaller than the net distribution. So, in this view, the variance stemming from the categories would be a very small contributor to the overall observed variance in the categories, at best. Furthermore, this move to accounting for Normality effectively concedes that categories with parametric Normal distributions are necessary. Consequently, both these related claims would run counter to the crucial claims of Exemplar theory, which is that the actual (not parameterized) distributions of the categories stored in memory as exemplars are what critically affect perception.

Along with the actual results presented in this paper, one also needs to consider the space of possibilities when assessing whether theories provide true explanations. It is worth highlighting here that a more complex model can always account for more data through over-fitting simply because it has more flexible parameters. With more flexible or richer theories, the same flexibility that allows one to accommodate observed patterns is also a curse when it comes to *explaining* them. This brings to fore the tension between explanations and descriptions/accounts. If our theories are richer in structures and possibilities, they necessarily become weaker explanatory theories. For example, if there was a sequential number generator that generates a sequence of 10 consecutive natural numbers with a little bit of noise. A 10 parameter model will perfectly fit the data, but a 2 parameter model that tracks the beginning and end of the sequence might not. From this, one shouldn't infer that the 10 parameter model is a reflection of reality. What really matters is how well the model explains the data or for that matter truly predicts unseen/untrained data (*i.e.*, when severely tested). The 10 parameter model would have been perfectly consistent with another random set of 10 numbers too, while the 2 parameter model is severely constrained and therefore is far less flexible. Similarly, observations of fit between very flexible models (for example, Exemplar theories) and extant data should be taken with some skepticism given this issue. Perhaps a more productive way to make progress is to have much smaller/simpler theories, and try to derive the bewildering complexity of the observed facts from interacting sub-systems or other cognitive systems as has been the driving force in modern work on linguistic competence (Chomsky 1965; Du and Durvasula 2025). However, further discussion of this issue takes us far beyond the scope of the current paper.

Table 5 lists the number of parameters for each of the theories, as studied in this paper. Unlike Bayesian

and Prototype theories, where the number of parameters increases linearly with the number of categories, for Category Boundary theories, the number of parameters increases quadratically with the number of categories due to the combinatoric function involved. However, this is not an issue as long as the number of categories is small and finite. In contrast, the number of parameters is inordinately large for Exemplar representations, not only because of the number of dimensions argued for by some in the literature, but because every single Exemplar has to be stored along all those phonetic dimensions. Consequently, Exemplar theories are far more complex and flexible than the other theories, and in such flexibility, one can see the issue with providing and explanation for the speech categorization laws that formed the basis of this paper. Relatedly, additional free parameters in theories should be seen with quite some skepticism as they enlarge the space of possibilities, and allow for easy curve-fitting.

Table 5: The number of parameters for each formal model of speech perception, as discussed in this paper

Model	Calculation	Total number of parameters
Prototype Perception	N means + distance noise/variance parameter	N+1
Category Boundary	C(N,2) means and C(N,2) variances	2*C(N,2)
Bayesian categorization	N means, N variances, N priors	3N
Exemplar Representation	Unspecified	Unspecified

Some have tried to solve the general issue of theory assessment through cross-validation, where a portion of the data is withheld for testing, and the process is repeated with different subsets of the data (Cutting 2000); however, even cross-validation techniques are quite complicated to learn from (Bates et al. 2024). While the method is able to test out-of-sample performance in a very narrow sense, it is still constrained by the fact that the unseen (test) data is drawn from the same population as the training data. What is really needed is a separation in the *type* or *patterns* between the data that the models are trained on, and what they are tested on; *i.e.*, what is needed for true evidentiary support is that the testing data be from a different population than the training data. Therefore, the question is whether theories predict *novel phenomena* once they have been trained on some data, not whether they can fit similar, but unseen, data well. In this regard, because data fitting is insufficient to establish the scientific value of a model/theory,

true severe testing (Mayo 1996; Mayo 2005; Peirce 1931, 1958) of many theories of perception is still needed. I raise this issue here primarily to highlight the problem, and to encourage future researchers to take up the challenge of severely testing computational models.

Before concluding, there are a few issues that need elaboration. First, in this paper, I did not assess “hybrid” theories that involve exemplar theories interacting with one of the other theories (Goldrick and Cole 2023; Pierrehumbert 2002, 2016). I made this choice because formal characterizations require precise theories of interaction, and these are generally missing for hybrid theories of speech perception/categorization. Furthermore, given that theories are at least as expressive/flexible as their most flexible part, such hybrid theories obtain all of the weaknesses of Exemplar theories particularly with respect to theoretical flexibility. Consequently, it is currently difficult to see how one can show that such theories have true explanations of the laws of speech categorization, restated in (3).

Second, throughout this paper, I have focussed on the *formal* properties of different theories of speech categorization. This is in the spirit of Marr (1982) who suggested that perhaps the computational level of analysis is the most important and primary to other analysis levels, and should precede analysis at other levels as it constrains the nature of the algorithmic and implementational levels. While it is not clear Marr (1982) intended it, in recent years it has been productive to think of algorithmic models as an approximation of the computational level description, where the algorithm converges (in the limit) to the computational level description. Such a view of the two levels has been very productive in understanding cognitive functions generally, in a variety of domains: distance estimation in *Cataglyphis* ants (Wehner and Müller 1993; Wehner 2003), area estimation in *Leptothorax Albipennis* ants (Mallon and Franks 2000; Mugford et al. 2001), localization in Barn owls (Bae and Peña 2024), exception learning in language (Yang 2016), amongst others. Using a similar view of the distinction between the computational and algorithmic levels of analyses of cognitive functions, some have argued that Exemplar theories can be thought of as approximate algorithmic implementations of Bayesian theories, which are often claimed to be computational-level theories in Marr’s sense of the terms (Feldman et al. 2009; Shi et al. 2008). However, to the extent that the space of identification functions under Exemplar theories is much larger than that under Bayesian theories, this connection is not possible to make without severe restrictions on what is a possible category distribution in

Exemplar representations. Generally, since Exemplar theories do not converge to Bayesian theories in the limit, the former cannot be thought of as an algorithmic instantiation of the latter. In fact, in Shi et al.'s (2008) simulation of the Exemplar theory, they model (without discussion) the relevant categories to be Normally distributed with equal variances. As shown in this paper, it is exactly in this case (and only in this case) that Exemplar theory is able to predict both the weak and strong laws of speech categorization. Therefore, the simulations in Shi et al. (2008) do not quite explore the full expressive power of Exemplar theories. Consequently, the approximate equivalence that they argue for between Bayesian theories and Exemplar theories is not generally correct.

Finally, the task of speech perception is much more complicated than the 2 category single phonetic dimension version I have presented in the paper. However, the issues I have raised for different theories are likely to aggregate and even compound with more categories and dimensions. Furthermore, a theory that is found to be problematic even in the simplest of settings is unlikely to scale up in more realistic scenarios. Minimally, I hope the work will engender more formal comparison of theories of categorization, and cognitive theories more generally.

A Appendix

A.1 Simplifying equation $\mathbf{IP}(Cat_1|x) = \mathbf{IP}(Cat_2|x)$

$$\frac{1}{1 + \frac{\sigma_2}{\sigma_1} * \frac{\mathbf{IP}(Cat_1)}{\mathbf{IP}(Cat_2)} * e^{\left(-\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2\right) - \left(-\frac{1}{2}\left(\frac{x-\mu_2}{\sigma_2}\right)^2\right)}} = \frac{1}{1 + \frac{\sigma_1}{\sigma_2} * \frac{\mathbf{IP}(Cat_2)}{\mathbf{IP}(Cat_1)} * e^{\left(-\frac{1}{2}\left(\frac{x-\mu_2}{\sigma_2}\right)^2\right) - \left(-\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2\right)}}$$

$$\Rightarrow \frac{1}{1 + \frac{\sigma_2}{\sigma_1} * \frac{\mathbf{IP}(Cat_1)}{\mathbf{IP}(Cat_2)} * e^{\left(-\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2\right) + \left(\frac{1}{2}\left(\frac{x-\mu_2}{\sigma_2}\right)^2\right)}} = \frac{1}{1 + \frac{\sigma_1}{\sigma_2} * \frac{\mathbf{IP}(Cat_2)}{\mathbf{IP}(Cat_1)} * e^{\left(-\frac{1}{2}\left(\frac{x-\mu_2}{\sigma_2}\right)^2\right) + \left(\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2\right)}}$$

Simplifying,

$$\Rightarrow \frac{\sigma_2}{\sigma_1} * \frac{\mathbf{IP}(Cat_1)}{\mathbf{IP}(Cat_2)} * e^{\left(-\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2\right) + \left(\frac{1}{2}\left(\frac{x-\mu_2}{\sigma_2}\right)^2\right)} = \frac{\sigma_1}{\sigma_2} * \frac{\mathbf{IP}(Cat_2)}{\mathbf{IP}(Cat_1)} * e^{\left(-\frac{1}{2}\left(\frac{x-\mu_2}{\sigma_2}\right)^2\right) + \left(\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2\right)}$$

multiplying both sides by $\frac{\sigma_1}{\sigma_1} * \frac{\mathbf{IP}(Cat_2)}{\mathbf{IP}(Cat_1)}$, assuming none of the terms are zero.

$$\Rightarrow e^{\left(-\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2\right) + \left(\frac{1}{2}\left(\frac{x-\mu_2}{\sigma_2}\right)^2\right)} = \left(\frac{\sigma_1}{\sigma_2} * \frac{\mathbb{P}(Cat_2)}{\mathbb{P}(Cat_1)}\right)^2 * e^{\left(-\frac{1}{2}\left(\frac{x-\mu_2}{\sigma_2}\right)^2\right) + \left(\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2\right)}$$

Taking logs on both sides. Note, $\log(x*y) = \log(x) + \log(y)$

$$\Rightarrow \left(-\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2\right) + \left(\frac{1}{2}\left(\frac{x-\mu_2}{\sigma_2}\right)^2\right) = \log\left(\frac{\sigma_1}{\sigma_2} * \frac{\mathbb{P}(Cat_2)}{\mathbb{P}(Cat_1)}\right)^2 + \left(-\frac{1}{2}\left(\frac{x-\mu_2}{\sigma_2}\right)^2\right) + \left(\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2\right)$$

Subtracting $\left(-\frac{1}{2}\left(\frac{x-\mu_2}{\sigma_2}\right)^2\right) + \left(\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2\right)$ from both sides,

$$\begin{aligned} \Rightarrow & \left(-\left(\frac{x-\mu_1}{\sigma_1}\right)^2\right) + \left(\left(\frac{x-\mu_2}{\sigma_2}\right)^2\right) = \log\left(\frac{\sigma_1}{\sigma_2} * \frac{\mathbb{P}(Cat_2)}{\mathbb{P}(Cat_1)}\right)^2 \\ \Rightarrow & \left(-\frac{x^2 + \mu_1^2 - 2*x*\mu_1}{\sigma_1^2}\right) + \left(\frac{x^2 + \mu_2^2 - 2*x*\mu_2}{\sigma_2^2}\right) = \log\left(\frac{\sigma_1}{\sigma_2} * \frac{\mathbb{P}(Cat_2)}{\mathbb{P}(Cat_1)}\right)^2 \end{aligned}$$

multiplying both sides by $\sigma_1^2 * \sigma_2^2$

$$\Rightarrow -\sigma_2^2 x^2 - \sigma_2^2 \mu_1^2 + \sigma_2^2 2x\mu_1 + \sigma_1^2 x^2 + \sigma_1^2 \mu_2^2 - \sigma_1^2 2x\mu_2 = \sigma_1^2 \sigma_2^2 \log\left(\frac{\sigma_1}{\sigma_2} * \frac{\mathbb{P}(Cat_2)}{\mathbb{P}(Cat_1)}\right)^2$$

Subtracting $-\sigma_2^2 \mu_1^2 + \sigma_1^2 \mu_2^2$ from both sides

$$\Rightarrow -\sigma_2^2 x^2 + \sigma_2^2 2x\mu_1 + \sigma_1^2 x^2 - \sigma_1^2 2x\mu_2 = \sigma_1^2 \sigma_2^2 \log\left(\frac{\sigma_1}{\sigma_2} * \frac{\mathbb{P}(Cat_2)}{\mathbb{P}(Cat_1)}\right)^2 + \sigma_2^2 \mu_1^2 - \sigma_1^2 \mu_2^2$$

$$\Rightarrow (\sigma_1^2 - \sigma_2^2)x^2 + 2x(\sigma_2^2 \mu_1 - \sigma_1^2 \mu_2) = \sigma_1^2 \sigma_2^2 \log\left(\frac{\sigma_1}{\sigma_2} * \frac{\mathbb{P}(Cat_2)}{\mathbb{P}(Cat_1)}\right)^2 + \sigma_2^2 \mu_1^2 - \sigma_1^2 \mu_2^2$$

To simplify it further, there are two cases,

Case 1: if $(\sigma_1^2 - \sigma_2^2) = 0$

$$2x(\sigma_2^2 \mu_1 - \sigma_1^2 \mu_2) = \sigma_1^2 \sigma_2^2 \log\left(\frac{\sigma_1}{\sigma_2} * \frac{\mathbb{P}(Cat_2)}{\mathbb{P}(Cat_1)}\right)^2 + \sigma_2^2 \mu_1^2 - \sigma_1^2 \mu_2^2$$

Diving both sides by $2 * (\sigma_2^2 \mu_1 - \sigma_1^2 \mu_2)$, assuming the term is $\neq 0$

$$\Rightarrow x = \frac{\sigma_1^2 \sigma_2^2 \log\left(\frac{\sigma_1}{\sigma_2} \frac{\text{IP}(\text{Cat}_2)}{\text{IP}(\text{Cat}_1)}\right)^2 + \sigma_2^2 \mu_1^2 - \sigma_1^2 \mu_2^2}{2 * (\sigma_2^2 \mu_1 - \sigma_1^2 \mu_2)}$$

The above suggests that when the variances are equal, there is only a single category boundary.

Case 2: if $(\sigma_1^2 - \sigma_2^2) \neq 0$

Dividing by $(\sigma_1^2 - \sigma_2^2)$ on both sides

$$\Rightarrow x^2 + 2x \left(\frac{\sigma_2^2 \mu_1 - \sigma_1^2 \mu_2}{\sigma_1^2 - \sigma_2^2} \right) = \frac{\sigma_1^2 \sigma_2^2 \log\left(\frac{\sigma_1}{\sigma_2} \frac{\text{IP}(\text{Cat}_2)}{\text{IP}(\text{Cat}_1)}\right)^2 + \sigma_2^2 \mu_1^2 - \sigma_1^2 \mu_2^2}{(\sigma_1^2 - \sigma_2^2)}$$

Adding $\left(\frac{\sigma_2^2 \mu_1 - \sigma_1^2 \mu_2}{\sigma_1^2 - \sigma_2^2} \right)^2$ on both sides to complete the square on the left hand side,

$$\begin{aligned} \Rightarrow x^2 + 2x \left(\frac{\sigma_2^2 \mu_1 - \sigma_1^2 \mu_2}{\sigma_1^2 - \sigma_2^2} \right) + \left(\frac{\sigma_2^2 \mu_1 - \sigma_1^2 \mu_2}{\sigma_1^2 - \sigma_2^2} \right)^2 &= \frac{\sigma_1^2 \sigma_2^2 \log\left(\frac{\sigma_1}{\sigma_2} \frac{\text{IP}(\text{Cat}_2)}{\text{IP}(\text{Cat}_1)}\right)^2 + \sigma_2^2 \mu_1^2 - \sigma_1^2 \mu_2^2}{(\sigma_1^2 - \sigma_2^2)} + \left(\frac{\sigma_2^2 \mu_1 - \sigma_1^2 \mu_2}{\sigma_1^2 - \sigma_2^2} \right)^2 \\ \Rightarrow \left(x + \left(\frac{\sigma_2^2 \mu_1 - \sigma_1^2 \mu_2}{\sigma_1^2 - \sigma_2^2} \right) \right)^2 &= \frac{\sigma_1^2 \sigma_2^2 \log\left(\frac{\sigma_1}{\sigma_2} \frac{\text{IP}(\text{Cat}_2)}{\text{IP}(\text{Cat}_1)}\right)^2 + \sigma_2^2 \mu_1^2 - \sigma_1^2 \mu_2^2}{(\sigma_1^2 - \sigma_2^2)} + \left(\frac{\sigma_2^2 \mu_1 - \sigma_1^2 \mu_2}{\sigma_1^2 - \sigma_2^2} \right)^2 \end{aligned}$$

Taking square root on both sides,

$$\Rightarrow x + \left(\frac{\sigma_2^2 \mu_1 - \sigma_1^2 \mu_2}{\sigma_1^2 - \sigma_2^2} \right) = \pm \sqrt{\frac{\sigma_1^2 \sigma_2^2 \log\left(\frac{\sigma_1}{\sigma_2} \frac{\text{IP}(\text{Cat}_2)}{\text{IP}(\text{Cat}_1)}\right)^2 + \sigma_2^2 \mu_1^2 - \sigma_1^2 \mu_2^2}{(\sigma_1^2 - \sigma_2^2)} + \left(\frac{\sigma_2^2 \mu_1 - \sigma_1^2 \mu_2}{\sigma_1^2 - \sigma_2^2} \right)^2}$$

Subtracting $\left(\frac{\sigma_2^2 \mu_1 - \sigma_1^2 \mu_2}{\sigma_1^2 - \sigma_2^2} \right)$ from both sides,

$$x = \pm \sqrt{\frac{\sigma_1^2 \sigma_2^2 \log\left(\frac{\sigma_1}{\sigma_2} \frac{\text{IP}(\text{Cat}_2)}{\text{IP}(\text{Cat}_1)}\right)^2 + \sigma_2^2 \mu_1^2 - \sigma_1^2 \mu_2^2}{(\sigma_1^2 - \sigma_2^2)} + \left(\frac{\sigma_2^2 \mu_1 - \sigma_1^2 \mu_2}{\sigma_1^2 - \sigma_2^2} \right)^2} - \left(\frac{\sigma_2^2 \mu_1 - \sigma_1^2 \mu_2}{\sigma_1^2 - \sigma_2^2} \right)$$

Re-arranging,

$$x = \left(\frac{\sigma_1^2 \mu_2 - \sigma_2^2 \mu_1}{\sigma_1^2 - \sigma_2^2} \right) \pm \sqrt{\frac{\sigma_1^2 \sigma_2^2 \log\left(\frac{\sigma_1}{\sigma_2} \frac{\text{IP}(\text{Cat}_2)}{\text{IP}(\text{Cat}_1)}\right)^2 + \sigma_2^2 \mu_1^2 - \sigma_1^2 \mu_2^2}{(\sigma_1^2 - \sigma_2^2)} + \left(\frac{\sigma_2^2 \mu_1 - \sigma_1^2 \mu_2}{\sigma_1^2 - \sigma_2^2} \right)^2}$$

The above suggests that every time the variances are un-equal, there are *two* categorical boundaries (or 50%

points).

A.2 Simplifying the first derivative of the identification function and solving

for x

Getting the first derivative of the identification function.

$$\begin{aligned}
 \frac{d}{dx}(\mathbb{P}(Cat_2|x)) &= \frac{d}{dx} \frac{1}{1 + e^{\left(-\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2\right) + \left(\frac{1}{2}\left(\frac{x-\mu_2}{\sigma_2}\right)^2\right) + \log\left(\frac{\sigma_2}{\sigma_1} * \frac{\mathbb{P}(Cat_1)}{\mathbb{P}(Cat_2)}\right)}} \\
 &= \frac{d}{dx} \frac{1}{1 + e^{\left(-\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2\right) + \left(\frac{1}{2}\left(\frac{x-\mu_2}{\sigma_2}\right)^2\right) + \log\left(\frac{\sigma_2}{\sigma_1} * \frac{\mathbb{P}(Cat_1)}{\mathbb{P}(Cat_2)}\right)}} \\
 &= -1 * \frac{e^{\left(-\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2\right) + \left(\frac{1}{2}\left(\frac{x-\mu_2}{\sigma_2}\right)^2\right) + \log\left(\frac{\sigma_2}{\sigma_1} * \frac{\mathbb{P}(Cat_1)}{\mathbb{P}(Cat_2)}\right)} * \left(-\left(\frac{x-\mu_1}{\sigma_1}\right) * \frac{1}{\sigma_1} + \left(\frac{x-\mu_2}{\sigma_2}\right) * \frac{1}{\sigma_2}\right)}{\left(1 + e^{\left(-\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2\right) + \left(\frac{1}{2}\left(\frac{x-\mu_2}{\sigma_2}\right)^2\right) + \log\left(\frac{\sigma_2}{\sigma_1} * \frac{\mathbb{P}(Cat_1)}{\mathbb{P}(Cat_2)}\right)}\right)^2}
 \end{aligned} \tag{38}$$

Now equating the derivative to 0, and then solving for x to find inflection points.

$$\begin{aligned}
 \frac{d}{dx}\mathbb{P}(Cat_2|x) &= -1 * \frac{e^{\left(-\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2\right) + \left(\frac{1}{2}\left(\frac{x-\mu_2}{\sigma_2}\right)^2\right) + \log\left(\frac{\sigma_2}{\sigma_1} * \frac{\mathbb{P}(Cat_1)}{\mathbb{P}(Cat_2)}\right)} * \left(-\left(\frac{x-\mu_1}{\sigma_1}\right) * \frac{1}{\sigma_1} + \left(\frac{x-\mu_2}{\sigma_2}\right) * \frac{1}{\sigma_2}\right)}{\left(1 + e^{\left(-\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2\right) + \left(\frac{1}{2}\left(\frac{x-\mu_2}{\sigma_2}\right)^2\right) + \log\left(\frac{\sigma_2}{\sigma_1} * \frac{\mathbb{P}(Cat_1)}{\mathbb{P}(Cat_2)}\right)}\right)^2} = 0 \\
 &\Rightarrow - \frac{e^{\left(-\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2\right) + \left(\frac{1}{2}\left(\frac{x-\mu_2}{\sigma_2}\right)^2\right) + \log\left(\frac{\sigma_2}{\sigma_1} * \frac{\mathbb{P}(Cat_1)}{\mathbb{P}(Cat_2)}\right)} * \left(\left(\frac{x-\mu_2}{\sigma_2}\right) - \left(\frac{x-\mu_1}{\sigma_1}\right)\right)}{\left(1 + e^{\left(-\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2\right) + \left(\frac{1}{2}\left(\frac{x-\mu_2}{\sigma_2}\right)^2\right) + \log\left(\frac{\sigma_2}{\sigma_1} * \frac{\mathbb{P}(Cat_1)}{\mathbb{P}(Cat_2)}\right)}\right)^2} = 0
 \end{aligned}$$

Using the minus sign to rearranging the terms inside the second multiplicand of the numerator,

$$\begin{aligned}
 &\Rightarrow \frac{e^{\left(-\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2\right) + \left(\frac{1}{2}\left(\frac{x-\mu_2}{\sigma_2}\right)^2\right) + \log\left(\frac{\sigma_2}{\sigma_1} * \frac{\mathbb{P}(Cat_1)}{\mathbb{P}(Cat_2)}\right)} * \left(\left(\frac{x-\mu_1}{\sigma_1}\right) - \left(\frac{x-\mu_2}{\sigma_2}\right)\right)}{\left(1 + e^{\left(-\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2\right) + \left(\frac{1}{2}\left(\frac{x-\mu_2}{\sigma_2}\right)^2\right) + \log\left(\frac{\sigma_2}{\sigma_1} * \frac{\mathbb{P}(Cat_1)}{\mathbb{P}(Cat_2)}\right)}\right)^2} = 0
 \end{aligned} \tag{39}$$

The denominator is necessarily positive. Therefore, one can multiply both sides by it and remove it.

Similarly, the exponential term in the numerator is necessarily non-zero for any finite set of parameter

815 values. Therefore, one can divide both sides by it and remove it.

816 Simplifying,

$$\Rightarrow \left(\frac{x - \mu_1}{\sigma_1^2} \right) - \left(\frac{x - \mu_2}{\sigma_2^2} \right) = 0$$

Assuming both variances are non-zero, simplifying,

$$\Rightarrow \left(\frac{\sigma_2^2}{\sigma_2^2} * \frac{x - \mu_1}{\sigma_1^2} \right) - \left(\frac{\sigma_1^2}{\sigma_1^2} * \frac{x - \mu_2}{\sigma_2^2} \right) = 0$$

Multiplying both sides by $\sigma_1^2 * \sigma_2^2$ and then simplifying,

$$\Rightarrow (\sigma_2^2 * x - \sigma_2^2 * \mu_2) - (\sigma_1^2 * x + \sigma_1^2 * \mu_1) = 0$$

Rearranging,

$$\Rightarrow x(\sigma_1^2 - \sigma_2^2) = \sigma_1^2 * \mu_2 - \sigma_2^2 * \mu_1$$

Assuming $\sigma_1 \neq \sigma_2$, dividing both sides by $(\sigma_1^2 - \sigma_2^2)$,

$$\Rightarrow x = \frac{\sigma_1^2 * \mu_2 - \sigma_2^2 * \mu_1}{\sigma_1^2 - \sigma_2^2}$$

817 The above result guarantees that whenever $\sigma_1 \neq \sigma_2$, there must be an inflection point such that the
 818 $\mathbb{P}(Cat_2|X = x)$ forms a u-shaped curve. Furthermore, when $\sigma_1 = \sigma_2$, there is no inflection point, and so we
 819 are able to retain the sigmoidal curve.

820 A.3 Establishing Convexity

821 Getting the second derivative of the identification function, using the first derivative derived in Eq. 38.

$$822 \quad \frac{d^2}{dx^2} \mathbb{P}(Cat_2|x) = \frac{d}{dx} \frac{e^{\left(-\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2\right) + \left(\frac{1}{2}\left(\frac{x-\mu_2}{\sigma_2}\right)^2\right) + \log\left(\frac{\sigma_2}{\sigma_1} * \frac{\mathbb{P}(Cat_1)}{\mathbb{P}(Cat_2)}\right)} * \left(\left(\frac{x-\mu_1}{\sigma_1^2}\right) - \left(\frac{x-\mu_2}{\sigma_2^2}\right)\right)}{\left(1 + e^{\left(-\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2\right) + \left(\frac{1}{2}\left(\frac{x-\mu_2}{\sigma_2}\right)^2\right) + \log\left(\frac{\sigma_2}{\sigma_1} * \frac{\mathbb{P}(Cat_1)}{\mathbb{P}(Cat_2)}\right)}\right)^2}$$

823

[illegible]

Removing the common term in the numerator of the second summand,

$$\begin{aligned}
&= \frac{\left(\frac{1}{\sigma_1^2} - \frac{1}{\sigma_2^2}\right)}{\left(2 + e^{\left(\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2\right) - \left(\frac{1}{2}\left(\frac{x-\mu_2}{\sigma_2}\right)^2\right) - \log\left(\frac{\sigma_2}{\sigma_1} * \frac{\text{IP}(Cat_1)}{\text{IP}(Cat_2)}\right)} + e^{\left(-\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2\right) + \left(\frac{1}{2}\left(\frac{x-\mu_2}{\sigma_2}\right)^2\right) + \log\left(\frac{\sigma_2}{\sigma_1} * \frac{\text{IP}(Cat_1)}{\text{IP}(Cat_2)}\right)}\right)} \\
&+ \left(\left(\frac{x-\mu_1}{\sigma_1^2}\right) - \left(\frac{x-\mu_2}{\sigma_2^2}\right)\right) * -1 * \left(\left(\frac{x-\mu_2}{\sigma_2^2}\right) - \left(\frac{x-\mu_1}{\sigma_1^2}\right)\right) * \\
&\left(\frac{e^{\left(\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2\right) - \left(\frac{1}{2}\left(\frac{x-\mu_2}{\sigma_2}\right)^2\right) - \log\left(\frac{\sigma_2}{\sigma_1} * \frac{\text{IP}(Cat_1)}{\text{IP}(Cat_2)}\right)} * -1 + e^{\left(-\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2\right) + \left(\frac{1}{2}\left(\frac{x-\mu_2}{\sigma_2}\right)^2\right) + \log\left(\frac{\sigma_2}{\sigma_1} * \frac{\text{IP}(Cat_1)}{\text{IP}(Cat_2)}\right)}}{\left(2 + e^{\left(\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2\right) - \left(\frac{1}{2}\left(\frac{x-\mu_2}{\sigma_2}\right)^2\right) - \log\left(\frac{\sigma_2}{\sigma_1} * \frac{\text{IP}(Cat_1)}{\text{IP}(Cat_2)}\right)} + e^{\left(-\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2\right) + \left(\frac{1}{2}\left(\frac{x-\mu_2}{\sigma_2}\right)^2\right) + \log\left(\frac{\sigma_2}{\sigma_1} * \frac{\text{IP}(Cat_1)}{\text{IP}(Cat_2)}\right)}\right)^2}\right)
\end{aligned}$$

Simplifying,

$$\begin{aligned}
&= \frac{\left(\frac{1}{\sigma_1^2} - \frac{1}{\sigma_2^2}\right)}{\left(2 + e^{\left(\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2\right) - \left(\frac{1}{2}\left(\frac{x-\mu_2}{\sigma_2}\right)^2\right) - \log\left(\frac{\sigma_2}{\sigma_1} * \frac{\text{IP}(Cat_1)}{\text{IP}(Cat_2)}\right)} + e^{\left(-\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2\right) + \left(\frac{1}{2}\left(\frac{x-\mu_2}{\sigma_2}\right)^2\right) + \log\left(\frac{\sigma_2}{\sigma_1} * \frac{\text{IP}(Cat_1)}{\text{IP}(Cat_2)}\right)}\right)} \\
&+ \left(\left(\frac{x-\mu_1}{\sigma_1^2}\right) - \left(\frac{x-\mu_2}{\sigma_2^2}\right)\right)^2 * \left(\frac{e^{\left(-\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2\right) + \left(\frac{1}{2}\left(\frac{x-\mu_2}{\sigma_2}\right)^2\right) + \log\left(\frac{\sigma_2}{\sigma_1} * \frac{\text{IP}(Cat_1)}{\text{IP}(Cat_2)}\right)} - e^{\left(\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2\right) - \left(\frac{1}{2}\left(\frac{x-\mu_2}{\sigma_2}\right)^2\right) - \log\left(\frac{\sigma_2}{\sigma_1} * \frac{\text{IP}(Cat_1)}{\text{IP}(Cat_2)}\right)}}{\left(2 + e^{\left(\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2\right) - \left(\frac{1}{2}\left(\frac{x-\mu_2}{\sigma_2}\right)^2\right) - \log\left(\frac{\sigma_2}{\sigma_1} * \frac{\text{IP}(Cat_1)}{\text{IP}(Cat_2)}\right)} + e^{\left(-\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2\right) + \left(\frac{1}{2}\left(\frac{x-\mu_2}{\sigma_2}\right)^2\right) + \log\left(\frac{\sigma_2}{\sigma_1} * \frac{\text{IP}(Cat_1)}{\text{IP}(Cat_2)}\right)}\right)}\right)
\end{aligned} \tag{40}$$

824 At $x = \frac{\sigma_1^2 * \mu_2 - \sigma_2^2 * \mu_1}{\sigma_1^2 - \sigma_2^2}$, the second summand of Eq. 40 becomes zero, since

825

$$\begin{aligned}
& \frac{x - \mu_2}{\sigma_2^2} - \frac{x - \mu_1}{\sigma_1^2} \\
&= \frac{\frac{\sigma_1^2 * \mu_2 - \sigma_2^2 * \mu_1}{\sigma_1^2 - \sigma_2^2} - \mu_2}{\sigma_2^2} - \frac{\frac{\sigma_1^2 * \mu_2 - \sigma_2^2 * \mu_1}{\sigma_1^2 - \sigma_2^2} - \mu_1}{\sigma_1^2} \\
&= \frac{\frac{\sigma_1^2 * \mu_2 - \sigma_2^2 * \mu_1 - \mu_2 * (\sigma_1^2 - \sigma_2^2)}{\sigma_1^2 - \sigma_2^2}}{\sigma_2^2} - \frac{\frac{\sigma_1^2 * \mu_2 - \sigma_2^2 * \mu_1 - \mu_1 * (\sigma_1^2 - \sigma_2^2)}{\sigma_1^2 - \sigma_2^2}}{\sigma_1^2} \\
&= \frac{\frac{-\sigma_2^2 * \mu_1 + \sigma_2^2 * \mu_2}{\sigma_1^2 - \sigma_2^2}}{\sigma_2^2} - \frac{\frac{\sigma_1^2 * \mu_2 - \sigma_1^2 * \mu_1}{\sigma_1^2 - \sigma_2^2}}{\sigma_1^2} \\
&= \frac{\frac{\sigma_2^2 * (\mu_2 - \mu_1)}{\sigma_1^2 - \sigma_2^2}}{\sigma_2^2} - \frac{\frac{\sigma_1^2 * (\mu_2 - \mu_1)}{\sigma_1^2 - \sigma_2^2}}{\sigma_1^2} \\
&= 0
\end{aligned}$$

826 Therefore, at $x = \frac{\sigma_1^2 * \mu_2 - \sigma_2^2 * \mu_1}{\sigma_1^2 - \sigma_2^2}$, the second derivative of the identification function is simply,

$$\begin{aligned}
& \frac{d^2}{d^2x} \mathbb{P} \left(Cat_2 | x = \frac{\sigma_1^2 * \mu_2 - \sigma_2^2 * \mu_1}{\sigma_1^2 - \sigma_2^2} \right) \\
&= \frac{\left(\frac{1}{\sigma_1^2} - \frac{1}{\sigma_2^2} \right)}{\left(2 + e^{\left(\frac{1}{2} \left(\frac{x - \mu_1}{\sigma_1} \right)^2 \right) - \left(\frac{1}{2} \left(\frac{x - \mu_2}{\sigma_2} \right)^2 \right) - \log \left(\frac{\sigma_2^2 * \mathbb{P}(Cat_1)}{\mathbb{P}(Cat_2)} \right)} + e^{\left(-\frac{1}{2} \left(\frac{x - \mu_1}{\sigma_1} \right)^2 \right) + \left(\frac{1}{2} \left(\frac{x - \mu_2}{\sigma_2} \right)^2 \right) + \log \left(\frac{\sigma_2^2 * \mathbb{P}(Cat_1)}{\mathbb{P}(Cat_2)} \right)} \right)}
\end{aligned} \tag{41}$$

827 The denominator in Eq. 41 is always positive, so the sign of the second derivative at the inflection point will
828 depend on the numerator.

$$\text{numerator of Eq. 41} = \left(\frac{1}{\sigma_1^2} - \frac{1}{\sigma_2^2} \right) = \frac{\sigma_2^2 - \sigma_1^2}{\sigma_1^2 * \sigma_2^2} \tag{42}$$

829 Therefore, when $\sigma_1^2 > \sigma_2^2$, the second derivative at the inflection point is positive; therefore, the identi-
830 cation function is convex. When $\sigma_1^2 < \sigma_2^2$, the second derivative at the inflection point is negative; therefore
831 the identification function is concave. And finally, when $\sigma_1^2 = \sigma_2^2$, the second derivative at the inflection
832 point is 0, therefore the categorization function is neither convex nor concave.

A.4 Testing the formal models on an actual phonetic dimension

In order to get a better understanding of the expected identification shapes under the four theories, it is useful to work with a real case of phonetic variation. Here, I will use category distributions related to voice onset times (VOT) of voiced and voiceless stops in American English. VOT is defined as the duration from release of a stop to the point at which voicing starts, for a variety of stop consonants in American English, and is one of the crucial acoustic cues that distinguishes voiced from voiceless stops (Abramson and Lisker 1970; Abramson and Whalen 2017; Chodroff et al. 2019; Chodroff and Wilson 2017, 2022; Lisker and Abramson 1964). This is particularly useful for our purposes as VOT is a single phonetic dimension (VOT) and it has been studied extensively in perception (Abramson and Lisker 1970; Kim et al. 2002; Kuhl and Miller 1978; Nearey and Rochet 1994; Nittrouer 2004).

Voiced stops in American English have VOTs close to 0 (or often even less than 0), while voiceless stops have VOTs that are much higher. The perceptual categorical boundary between the two categories has been observed to be around 25–35 ms (Abramson and Lisker 1970; Nearey and Rochet 1994). Chodroff and Wilson (2017) ran a production experiment focussed on the distributions of VOTs for different voiced and voiceless stops in American English. They recorded 24 American English speakers producing C_1VC_2 syllables presented orthographically in the carrier phrase “Say ___ again.”, where C_1 was one of 6 consonants [p^h t^h k^h b d g], V was one of 10 vowels [i ɪ eɪ ε æ ʌ ɑ ɔ uʊ u], and C_2 was one of [t d]. Participants completed 5 blocks with each syllable appearing once per block. The VOTs were manually marked for each word-initial stop, and defined as “the duration of the interval from the beginning of the stop release to the start of periodicity in the waveform or a visible f0 track” (p. 33). Therefore, all the VOT values were necessarily positive in their measurements. In what follows, I will look at the VOT values of the bilabial stops ([b] - voiced, [p^h] - voiceless) from Chodroff and Wilson (2017). The data was downloaded from the Open Science Foundation repository available at <https://osf.io/k6djr/>. The expected distribution of VOT values for each category are clearly observed in the data.

The estimated parameter values are in Table 6. Interestingly, the means and the modes of the two categories are quite close; therefore, it won’t make a substantial difference to choose one over the other in

our modelling of the prototypes of the categories.⁹

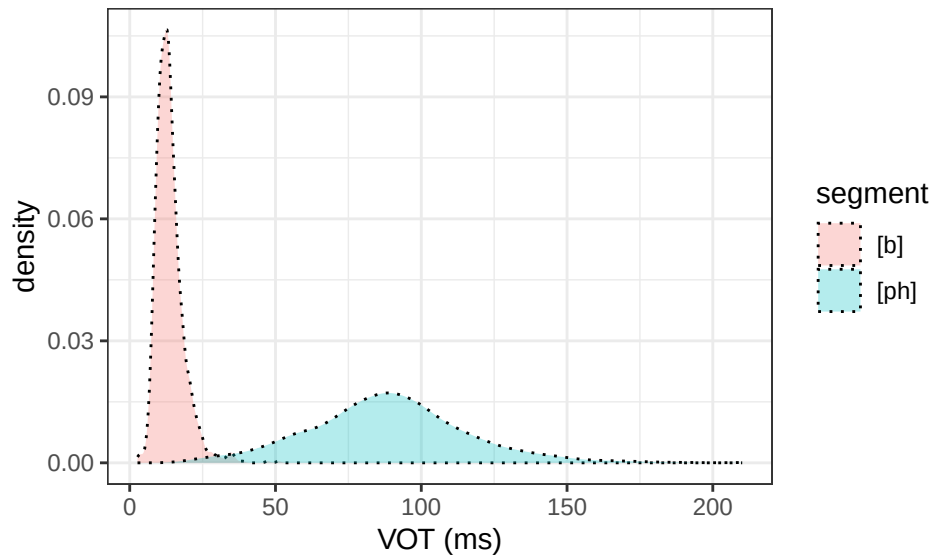


Figure 11: The distribution of VOTs for the two categories /p^h/, b/

Table 6: Parameter estimates (in ms) for each category

segment	meanVOT	modeVOT	sdVOT
[ph]	88.6	88.5	26.8
[b]	13.8	12.7	5.0

For the purposes of the testing, I will assume the both the categories [b p^h] are equiprobable in English, and therefore the priors/bias of the categories can be equal. One could debate the equal prior assumption, but it is not clear what frequencies to use to estimate such prior probabilities. Furthermore, the equal prior probability assumption is not unreasonable for the two categories [b p] when we look at either the overall or word-initial distributional frequencies in the CMU Dictionary (Weide 1994). Note, “overall” frequency here is defined as the number of such segments in word-initial and in pre-stressed positions, which are the environment in which aspiration is consistently observed in American English (Beckman et al. 2013; Iverson and Salmons 1995).

⁹Rosch (1978) very briefly mentions the possibility of using modes instead of means to represent categories.

	[p ^h]		[b]	
	Frequency (# words)	Est. Prob.	Frequency (# words)	Est. Prob.
Overall	9234	0.07	11092	0.08
Word-initial	7824	0.06	9628	0.07

Table 7: Frequencies/Estimated probabilities for /b, p^h/ from the CMU Dictionary (N = 133223 words) (Weide 1994)

Following Feldman (2015), Feldman et al. (2009), Kleinschmidt and Jaeger (2015), and Kronrod et al. (2016), if we make the assumption that listeners are learning the distributions from the production data they hear, then we could use production data to estimate the expected identification functions. In what follows, I will present the expected identification functions for all the theories in the interest of completeness. For Bayesian and Exemplar theories, we have the necessary information to get the identification function. However, for Categorical Boundary theories, we do not independently have the categorical boundary value, or the expected variance in the categorical boundary. Similarly, for the Prototype theory, we do not have the noise/variance in the distance estimation. Below, I will use some reasonable values, but these values don't affect the *shape* of the identification function, since in both cases the identification function is proved to be of a certain shape.

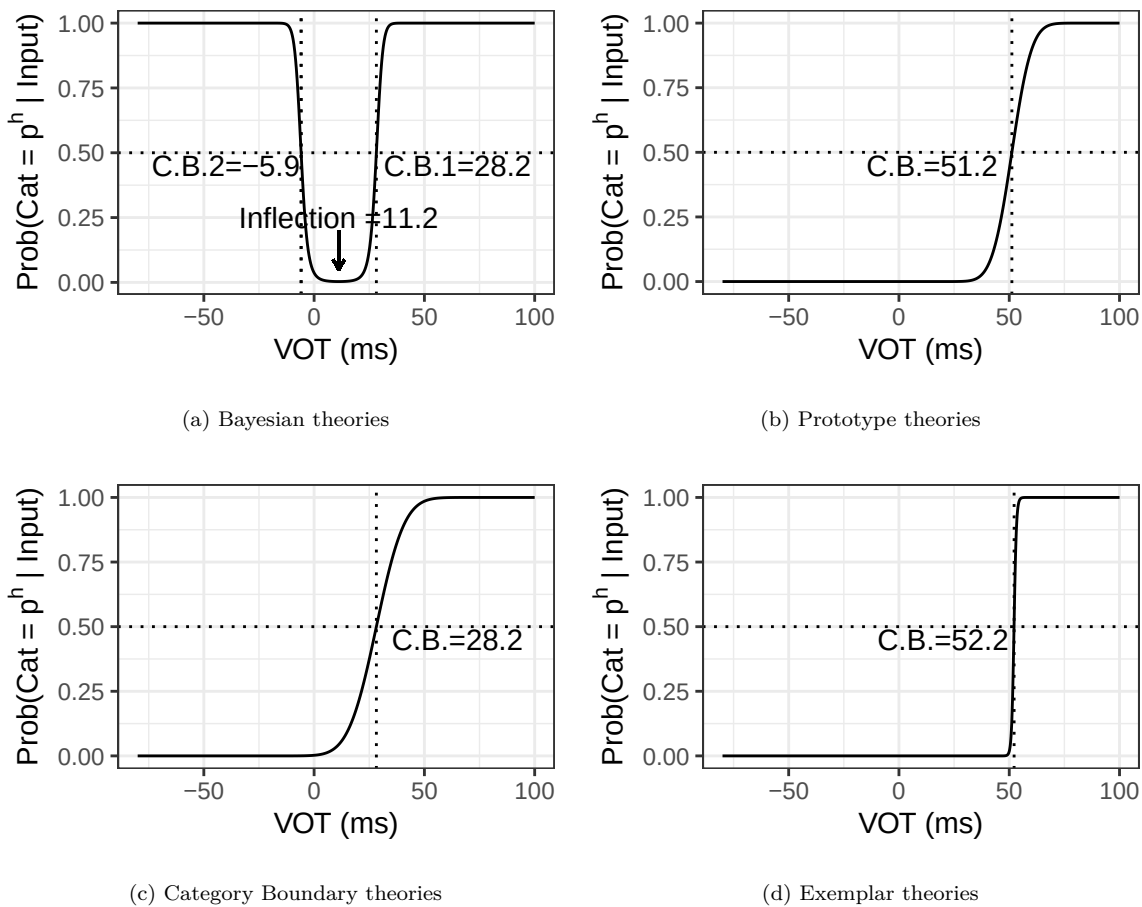


Figure 12: Identification functions for different theories of speech perception [C.B. = Categorical boundary in ms].

As can be seen from Figure 12, Bayesian theories predicts two categorical boundaries and consequently a convex identification function with an inflection point (minima) at 11.2 ms. Furthermore, one of the categorical boundaries (28.2 ms) is close to the observed VOT boundary of 30 ms, while the other (-5.9 ms) has not been observed in the literature. In contrast, the other three theories correctly result in sigmoidal shapes.

An important caveat is necessary in interpreting the above identification functions. The point of the simulation on real data was primarily to compare the *shapes* of the identification functions, and not much should be made about the fit or lack thereof between the predicted categorical boundaries and those observed in actual experiments. This is because none of the theories discussed have clear claims of how the relevant parameters are estimated — as discussed in the Introduction, the issue of learning is separate from the issue

of predicted identification function shapes. For example, in unstressed syllables, the VOTs of /b p/ are lower than in stressed or word-initial positions. So, if those distributions were included to the full distribution of /b p/ in the modelling, then all the models would predict lower categorical boundaries. In fact, under those circumstances, it is possible that Bayesian categorization theories would predict both the categorical boundaries to be below the observed range of 25–35 ms, while the other theories predict those within the correct range.

References

- Abramson, Arthur S and Leigh Lisker (1970). “Discriminability along the voicing continuum: Cross-language tests.” In: *Proceedings of the 6th International Congress of Phonetic Sciences*. Vol. 196. 7. Academia Prague, pp. 569–573.
- Abramson, Arthur S and Douglas H Whalen (2017). “Voice Onset Time (VOT) at 50: Theoretical and practical issues in measuring voicing distinctions.” In: *Journal of phonetics* 63, pp. 75–86.
- Apfelbaum, Keith S, Ethan Kutlu, Bob McMurray, and Efthymia C Kapnoula (2022). “Don’t force it! Gradient speech categorization calls for continuous categorization tasks.” In: *The Journal of the Acoustical Society of America* 152.6, pp. 3728–3745.
- Bae, Andrea and Jose L Peña (2024). “Barn owls specialized sound-driven behavior: Lessons in optimal processing and coding by the auditory system.” In: *Hearing Research* 443, p. 108952. ISSN: 0378-5955. DOI: <https://doi.org/10.1016/j.heares.2024.108952>. URL: <https://www.sciencedirect.com/science/article/pii/S0378595524000054>.
- Bates, Stephen, Trevor Hastie, and Robert Tibshirani (2024). “Cross-validation: what does it estimate and how well does it do it?” In: *Journal of the American Statistical Association* 119.546, pp. 1434–1445.
- Beare, Aleeza Cerf (1963). “Color-name as a function of wave-length.” In: *The American Journal of Psychology* 76.2, pp. 248–256.
- Beckman, Jill, Michael Jessen, and Catherine Ringen (2013). “Empirical evidence for laryngeal features: Aspirating vs. true voice languages.” In: *Journal of Linguistics* 49.2, pp. 259–284.

- Bertelson, Paul, Jean Vroomen, and Béatrice De Gelder (2003). "Visual recalibration of auditory speech identification: a McGurk aftereffect." In: *Psychological Science* 14.6, pp. 592–597.
- Bornstein, Marc H and Nancy O Korda (1985). "Identification and adaptation of hue: Parallels in the operation of mechanisms that underlie categorical perception in vision and in audition." In: *Psychological Research* 47.1, pp. 1–17.
- Bornstein, MH (1987). "Perceptual categories in vision and audition." In: *Categorical perception, the ground-work for cognition*. Cambridge, MA, USA: Cambridge University Press, pp. 287–300.
- Caplan, Spencer and Karthik Durvasula (under review). "The Discrete Perception Of Continuous Speech." In: URL: https://osf.io/preprints/psyarxiv/s4ch3_v1.
- Caplan, Spencer, Alon Hafri, and John C. Trueswell (2021). "Now You Hear Me, Later You Don't: The Immediacy of Linguistic Computation and the Representation of Speech." In: *Psychological Science* 32.3. PMID: 33617735, pp. 410–423. DOI: [10.1177/0956797620968787](https://doi.org/10.1177/0956797620968787). eprint: <https://doi.org/10.1177/0956797620968787>. URL: <https://doi.org/10.1177/0956797620968787>.
- Cartwright, Nancy (1983). *How the laws of physics lie*. OUP Oxford.
- Chodroff, Eleanor, Alessandra Golden, and Colin Wilson (2019). "Covariation of stop voice onset time across languages: Evidence for a universal constraint on phonetic realization." In: *The Journal of the Acoustical Society of America* 145.1, EL109–EL115.
- Chodroff, Eleanor and Colin Wilson (2017). "Structure in talker-specific phonetic realization: Covariation of stop consonant VOT in American English." In: *Journal of Phonetics* 61, pp. 30–47. ISSN: 0095-4470. DOI: <https://doi.org/10.1016/j.wocn.2017.01.001>. URL: <https://www.sciencedirect.com/science/article/pii/S0095447017300013>.
- Chodroff, Eleanor and Colin Wilson (2022). "Uniformity in phonetic realization: Evidence from sibilant place of articulation in American English." In: *Language* 98.2, pp. 250–289.
- Chomsky, Noam (1959). "On certain formal properties of grammars." In: *Information and Control* 2.2, pp. 137–167. ISSN: 0019-9958. DOI: [https://doi.org/10.1016/S0019-9958\(59\)90362-6](https://doi.org/10.1016/S0019-9958(59)90362-6). URL: <https://www.sciencedirect.com/science/article/pii/S0019995859903626>.
- Chomsky, Noam (1965). *Aspects of the Theory of Syntax*. Cambridge: MIT Press.

Chomsky, Noam and Andrea Moro (2022). *The secrets of words*. MIT Press.

Chomsky, Noam, Ian Roberts, and Jeffrey Watumull (2023). “Noam chomsky: The false promise of chatgpt.”

In: *The New York Times* 8.

Cohen, Henri and Claire Lefebvre (2005). *Handbook of categorization in cognitive science*. Elsevier.

Cummins, Robert (2000). ““How does it work?” vs. “What are the laws?” Two conceptions of psychological explanation.” In: ed. by F. Keil and R. Wilson, pp. 117–145.

Cutting, James E. (2000). “Accuracy, Scope, and Flexibility of Models.” In: *Journal of Mathematical Psychology* 44.1, pp. 3–19. ISSN: 0022-2496. DOI: <https://doi.org/10.1006/jmps.1999.1274>. URL: <https://www.sciencedirect.com/science/article/pii/S0022249699912749>.

Dong, Chenjie, Zhengye Wang, Ruqin Li, Uta Noppeney, and Suiping Wang (2025). “Variations in unisensory speech perception explain interindividual differences in McGurk illusion susceptibility.” In: *Psychonomic Bulletin & Review*, pp. 1–16.

Du, Naiyan and Karthik Durvasula (2025). “Psycholinguistics and Phonology: the forgotten foundations of generative phonology.” In: *Cambridge Elements on Phonology*.

Eimas, Peter D., William E. Cooper, and John D. Corbit (1973). “Some properties of linguistic feature detectors.” In: *Perception & Psychophysics* 13.2, pp. 247–252.

Eimas, Peter D. and John D. Corbit (1973). “Selective adaptation of linguistic feature detectors.” In: *Cognitive Psychology* 4.1, pp. 99–109. ISSN: 0010-0285. DOI: [https://doi.org/10.1016/0010-0285\(73\)90006-6](https://doi.org/10.1016/0010-0285(73)90006-6). URL: <http://www.sciencedirect.com/science/article/pii/0010028573900066>.

Eisner, Frank and James M McQueen (2005). “The specificity of perceptual learning in speech processing.” In: *Attention, Perception, & Psychophysics* 67.2, pp. 224–238.

Estes, William K (1956). “The problem of inference from curves based on group data.” In: *Psychological Bulletin* 53.2, pp. 134–140.

Feldman, Jacob (2015). “Bayesian models of perception: a tutorial introduction.” In: *The Oxford Handbook of Perceptual Organization*, pp. 1008–1026.

Feldman, Naomi H and Thomas L Griffiths (2007). “A Rational Account of the Perceptual Magnet Effect.” In: *Proceedings of the 29th Annual Conference of the Cognitive Science Society*. Austin, TX, pp. 257–262.

- Feldman, Naomi H, Thomas L Griffiths, and James L Morgan (2009). "The influence of categories on perception: explaining the perceptual magnet effect as optimal statistical inference." In: *Psychological review* 116.4, p. 752.
- Fry, D. B., Arthur S. Abramson, Peter D. Eimas, and Alvin M. Liberman (1962). "The Identification and Discrimination of Synthetic Vowels." In: *Language and Speech* 5.4, pp. 171–189. DOI: [10.1177/002383096200500401](https://doi.org/10.1177/002383096200500401). eprint: <https://doi.org/10.1177/002383096200500401>. URL: <https://doi.org/10.1177/002383096200500401>.
- Ganong, William F. (1980). "Phonetic categorization in auditory word perception." In: *Journal of Experimental Psychology: Human Perception and Performance* 6.1, p. 110.
- Goldinger, Stephen D. (1996). "Words and voices: Episodic traces in spoken word identification and recognition memory." In: *Journal of Experimental Psychology: Learning Memory and Cognition* 22.5. Cited by: 544, pp. 1166 – 1183. DOI: [10.1037/0278-7393.22.5.1166](https://doi.org/10.1037/0278-7393.22.5.1166). URL: <https://www.scopus.com/inward/record.uri?eid=2-s2.0-0030237557&doi=10.1037%2f0278-7393.22.5.1166&partnerID=40&md5=85b27cd9763a6f02d6c9405191f63af6>.
- Goldinger, Stephen D. (1997). "Words and voices: Perception and production in an episodic lexicon." In: ed. by Keith Johnson and John W. Mullenix. San Diego: Academic Press, pp. 33–66.
- Goldinger, Stephen D. (1998). "Echoes of Echoes? An Episodic Theory of Lexical Access." In: *Psychological Review* 105.2. Cited by: 1035, pp. 251 – 279. DOI: [10.1037/0033-295X.105.2.251](https://doi.org/10.1037/0033-295X.105.2.251). URL: <https://www.scopus.com/inward/record.uri?eid=2-s2.0-0032041690&doi=10.1037%2f0033-295X.105.2.251&partnerID=40&md5=8ebf7b1a56e4f5d585712a4da60eeef>.
- Goldrick, Matthew and Jennifer Cole (2023). "Advancement of phonetics in the 21st century: Exemplar models of speech production." In: *Journal of Phonetics* 99, p. 101254. ISSN: 0095-4470. DOI: <https://doi.org/10.1016/j.wocn.2023.101254>. URL: <https://www.sciencedirect.com/science/article/pii/S0095447023000438>.
- Guilford, Joy Paul (1954). *Psychometric methods*. McGraw-Hill.
- Harnad, Stevan (1987). *Categorical perception, the groundwork for cognition*. Ed. by Stevan Harnad. Cambridge, MA, USA: Cambridge University Press.

- Hay, Jennifer, Ryan Podlubny, Katie Drager, and Megan McAuliffe (2017). "Car-talk: Location-specific speech production and perception." In: *Journal of Phonetics* 65, pp. 94–109. ISSN: 0095-4470. DOI: <https://doi.org/10.1016/j.wocn.2017.06.005>. URL: <https://www.sciencedirect.com/science/article/pii/S0095447016300857>.
- Hayes, Bruce (2022). "Deriving the wug-shaped curve: A criterion for assessing formal theories of linguistic variation." In: *Annual Review of Linguistics* 8, pp. 473–494.
- Heffner, Christopher C, William J Idsardi, and Rochelle S Newman (2019). "Constraints on learning disjunctive, unidimensional auditory and phonetic categories." In: *Attention, Perception, & Psychophysics* 81.4, pp. 958–980.
- Hempel, Carl G (1942). "The function of general laws in history." In: *The journal of philosophy* 39.2, pp. 35–48.
- Hempel, Carl G and Paul Oppenheim (1948). "Studies in the Logic of Explanation." In: *Philosophy of science* 15.2, pp. 135–175.
- Hessen, A.J. van and M.E.H. Schouten (July 1999). "Categorical Perception as a Function of Stimulus Quality." In: *Phonetica* 56.1-2, pp. 56–72. ISSN: 0031-8388. DOI: [10.1159/000028441](https://doi.org/10.1159/000028441). eprint: <https://karger.com/pho/article-pdf/56/1-2/56/3434030/000028441.pdf>. URL: <https://doi.org/10.1159/000028441>.
- Huette, Stephanie (2012). "Dynamics of Color Category Formation and Boundaries." In: *Encyclopedia of Color Science and Technology*. Springer, pp. 1–5.
- Iverson, Gregory K and Joseph C Salmons (1995). "Aspiration and laryngeal representation in Germanic." In: *Phonology* 12, pp. 369–396.
- Johnson, Keith (1997). "Speech perception without speaker normalisation." In: ed. by Keith Johnson and John W. Mullennix. San Diego: Academic Press, pp. 145–165.
- Johnson, Keith and John W. Mullennix, eds. (1997). *Talker variability in speech processing*. San Diego: Academic Press.
- Kersten, Daniel and Alan Yuille (2003). "Bayesian models of object perception." In: *Current opinion in neurobiology* 13.2, pp. 150–158.

- Kim, Mi-Ryoung, Patrice Speeter Beddor, and Julie Horrocks (2002). "The contribution of consonantal and vocalic information to the perception of Korean initial stops." In: *Journal of Phonetics* 30.1, pp. 77–100.
- Kitcher, Philip (1989). "Explanatory unification and the causal structure of the world." In: *Minnesota studies in the philosophy of science, volume XIII: Scientific explanation*. Ed. by Phillip Kitcher and Wesley Salmon. Minneapolis, MN, USA: University of Minnesota Press, pp. 410–505.
- Kleinschmidt, Dave F. and T. Florian Jaeger (2015). "Robust speech perception: Recognize the familiar, generalize to the similar, and adapt to the novel." In: *Psychological Review* 122.2, pp. 148–203.
- Kraljic, Tanya and Arthur G Samuel (2005). "Perceptual learning for speech: Is there a return to normal?" In: *Cognitive psychology* 51.2, pp. 141–178.
- Kroch, Anthony S (1989). "Reflexes of grammar in patterns of language change." In: *Language variation and change* 1.3, pp. 199–244.
- Kronrod, Yakov, Emily Coppess, and Naomi H Feldman (2016). "A unified account of categorical effects in phonetic perception." In: *Psychonomic bulletin & review* 23.6, pp. 1681–1712.
- Kuhl, Patricia K (1991). "Human adults and human infants show a "perceptual magnet effect" for the prototypes of speech categories, monkeys do not." In: *Perception & psychophysics* 50.2, pp. 93–107.
- Kuhl, Patricia K. and James D. Miller (1978). "Speech perception by the chinchilla: Identification functions for synthetic VOT stimuli." In: *The Journal of the Acoustical Society of America* 63.3, pp. 905–917. ISSN: 0001-4966. DOI: [10.1121/1.381770](https://doi.org/10.1121/1.381770). eprint: https://pubs.aip.org/asa/jasa/article-pdf/63/3/905/11967667/905_1_online.pdf. URL: <https://doi.org/10.1121/1.381770>.
- Labov, William (2001). *Principles of Linguistic Change. Volume 2: Social factors*. Vol. 2. Oxford, UK: Blackwell.
- Lakatos, Imre (1970). "Falsification and the Methodology of Scientific Research Programmes." In: *Criticism and the Growth of Knowledge: Proceedings of the International Colloquium in the Philosophy of Science, London, 1965*. Ed. by Imre Lakatos and Alan Musgrave. Vol. 4. Cambridge University Press, pp. 91–196. DOI: [10.1017/CB09781139171434.009](https://doi.org/10.1017/CB09781139171434.009).
- Liberman, Alvin M, Franklin S Cooper, Donald P Shankweiler, and Michael Studdert-Kennedy (1967). "Perception of the speech code." In: *Psychological review* 74.6, p. 431.

- Liberman, Alvin M, Katherine Safford Harris, Howard S Hoffman, and Belver C Griffith (1957). "The discrimination of speech sounds within and across phoneme boundaries." In: *Journal of experimental psychology* 54.5, p. 358.
- Limpert, Eckhard, Werner A Stahel, and Markus Abbt (2001). "Log-normal distributions across the sciences: keys and clues: on the charms of statistics, and how mechanical models resembling gambling machines offer a link to a handy way to characterize log-normal distributions, which can provide deeper insight into variability and probability—normal or log-normal: that is the question." In: *BioScience* 51.5, pp. 341–352.
- Lisker, Leigh and Arthur S. Abramson (1964). "A Cross-Language Study of Voicing in Initial Stops: Acoustical Measurements." In: *WORD* 20.3, pp. 384–422. DOI: [10.1080/00437956.1964.11659830](https://doi.org/10.1080/00437956.1964.11659830). eprint: <https://doi.org/10.1080/00437956.1964.11659830>. URL: <https://doi.org/10.1080/00437956.1964.11659830>.
- Lotto, Andrew J, Keith R Kluender, and Lori L Holt (1998). "Depolarizing the perceptual magnet effect." In: *The Journal of the Acoustical Society of America* 103.6, pp. 3648–3655.
- Lyon, Aidan (2014). "Why are normal distributions normal?" In: *The British Journal for the Philosophy of Science*.
- Mallon, Eamonn B. and Nigel R. Franks (2000). "Ants Estimate Area Using Buffon's Needle." In: *Proceedings: Biological Sciences* 267.1445, pp. 765–770. ISSN: 09628452, 14712954. URL: <http://www.jstor.org/stable/2665875> (visited on 11/18/2025).
- Marr, David (1982). *Vision: A Computational approach*. San Francisco: Freeman & Co.
- Massaro, Dominic W and Michael M Cohen (1983). "Categorical or continuous speech perception: A new test." In: *Speech communication* 2.1, pp. 15–35.
- Massaro, Dominic W and Gregg C Oden (1980). "Evaluation and integration of acoustic features in speech perception." In: *The Journal of the Acoustical Society of America* 67.3, pp. 996–1013.
- Massaro, Dominic W and Jeffrey A Simpson (1987). *Speech perception by ear and eye: A paradigm for psychological inquiry*. Hillsdale, NJ, USA: Erlbaum.
- Mayo, Deborah G. (1996). *Error and the Growth of Experimental Knowledge*. University of Chicago Press.

- Mayo, Deborah G (2005). "Peircean induction and the error-correcting thesis." In: *Transactions of the Charles S. Peirce Society: A Quarterly Journal in American Philosophy* 41.2, pp. 299–319.
- McGurk, Harry and John MacDonald (1976). "Hearing lips and seeing voices." In: *Nature* 264.5588, pp. 746–748.
- McMurray, Bob (2022). "The myth of categorical perception." In: *The Journal of the Acoustical Society of America* 152.6, pp. 3819–3842.
- McMurray, Bob, Richard N Aslin, Michael K Tanenhaus, Michael J Spivey, and Dana Subik (2008). "Gradient sensitivity to within-category variation in words and syllables." In: *Journal of Experimental Psychology: Human Perception and Performance* 34.6, p. 1609.
- Miller, George A. (1957). "Some Effects of Intermittent Silence." In: *The American Journal of Psychology* 70.2, pp. 311–314. ISSN: 00029556. URL: <http://www.jstor.org/stable/1419346> (visited on 10/10/2025).
- Miller, Joshua B and Andrew Gelman (2020). "Laplace's theories of cognitive illusions, heuristics and biases." In: *Statistical science* 35.2, pp. 159–170.
- Mugford, S. T., E. B. Mallon, and N. R. Franks (Nov. 2001). "The accuracy of Buffon's needle: a rule of thumb used by ants to estimate area." In: *Behavioral Ecology* 12.6, pp. 655–658. ISSN: 1045-2249. DOI: [10.1093/beheco/12.6.655](https://doi.org/10.1093/beheco/12.6.655). eprint: <https://academic.oup.com/beheco/article-pdf/12/6/655/9742133/bhec-12-6-655.pdf>. URL: <https://doi.org/10.1093/beheco/12.6.655>.
- Nearey, Terrance M and Bernard L Rochet (1994). "Effects of place of articulation and vowel context on VOT production and perception for French and English stops." In: *Journal of the International Phonetic Association* 24.1, pp. 1–18.
- Nelson, Scott and Karthik Durvasula (2021). "Lexically-guided perceptual learning does generalize to new phonetic contexts." In: *Journal of Phonetics* 84, p. 101019.
- Nittrouer, Susan (2004). "The role of temporal and dynamic signal components in the perception of syllable-final stop voicing by children and adults." In: *The Journal of the Acoustical Society of America* 115.4, pp. 1777–1790.
- Norris, Dennis and James M. McQueen (2008). "Shortlist B: a Bayesian model of continuous speech recognition." In: *Psychological Review* 115.2, pp. 357–395.

- Norris, Dennis, James M. McQueen, and Anne Cutler (2003). "Perceptual learning in speech." In: *Cognitive Psychology* 30.2, pp. 1113–1126.
- Nosofsky, Robert M (1987). "Attention and learning processes in the identification and categorization of integral stimuli." In: *Journal of Experimental Psychology: Learning, Memory, and Cognition* 13.1, pp. 87–108.
- Nosofsky, Robert M (1988a). "Exemplar-based accounts of relations between classification, recognition, and typicality." In: *Journal of Experimental Psychology: learning, memory, and cognition* 14.4, p. 700.
- Nosofsky, Robert M (1988b). "Similarity, frequency, and category representations." In: *Journal of Experimental Psychology: learning, memory, and cognition* 14.1, p. 54.
- Oden, GC (1981). "A fuzzy propositional model of concept structure and use: A case study in object identification." In: *Applied systems and cybernetics (Vol. VI)*. Ed. by G. W. Lasker. Elmsford, NY, USA, pp. 2890–2897.
- Oden, Gregg C (1984). "Dependence, independence, and emergence of word features." In: *Journal of Experimental Psychology: Human Perception and Performance* 10.3, pp. 394–405.
- Oden, Gregg C and Dominic W Massaro (1978). "Integration of featural information in speech perception." In: *Psychological review* 85.3, p. 172.
- Peirce, Charles Sanders (1931). *Collected papers*. Ed. by C. Hartshorne and P. Weiss. Vol. I–VI. Cambridge, MA, USA: Harvard University Press.
- Peirce, Charles Sanders (1958). *Collected papers*. Ed. by A. Burks. Vol. VII–VIII. Cambridge, MA, USA: Harvard University Press.
- Piantadosi, Steven (2023). *Modern language models refute Chomsky's approach to language*. Paper retrieved from <https://lingbuzz.net/lingbuzz/007180>.
- Pierrehumbert, Janet B. (2002). "Word-specific phonetics." In: ed. by Carlos Gussenhoven and Natasha Warner. Berlin, New York: De Gruyter Mouton, pp. 101–140.
- Pierrehumbert, Janet B. (2016). "Phonological Representation: Beyond Abstract Versus Episodic." In: *Annual Review of Linguistics* 2.1, pp. 33–52. DOI: [10.1146/annurev-linguistics-030514-125050](https://doi.org/10.1146/annurev-linguistics-030514-125050). eprint:

<https://doi.org/10.1146/annurev-linguistics-030514-125050>. URL: <https://doi.org/10.1146/annurev-linguistics-030514-125050>.

Pisoni, David B (1973). "Auditory and phonetic memory codes in the discrimination of consonants and vowels." In: *Perception & psychophysics* 13.2, pp. 253–260.

Pisoni, David B (1975). "Auditory short-term memory and vowel perception." In: *Memory & Cognition* 3.1, pp. 7–18.

Pisoni, David B and Joan House Lazarus (1974). "Categorical and noncategorical modes of speech perception along the voicing continuum." In: *The Journal of the Acoustical Society of America* 55.2, pp. 328–333.

Port, Robert F. (2007). "The graphical basis of phones and phonemes." In: Amsterdam: John Benjamins, pp. 349–365.

Rosch, Eleanor (1977). "Human Categorization." In: *Advances in Cross-Cultural Psychology* 1. Ed. by Neil Warren. London: Academic Press, pp. 1–72.

Rosch, Eleanor (1978). "Principles of categorization." In: *Cognition and categorization*. Routledge, pp. 27–48.

Rosch, Eleanor and Carolyn B Mervis (1975). "Family resemblances: Studies in the internal structure of categories." In: *Cognitive psychology* 7.4, pp. 573–605.

Rysling, A., J. Kingston, A. Staub, A. Cohen, and J. Starns (2015). "Early Ganong effects." In: *Proceedings of the 19th International Congress of Phonetic Sciences*.

Shepard, Roger N (1957). "Stimulus and response generalization: A stochastic model relating generalization to distance in psychological space." In: *Psychometrika* 22.4, pp. 325–345.

Shepard, Roger N (1958). "Stimulus and response generalization: deduction of the generalization gradient from a trace model." In: *Psychological Review* 65.4, pp. 242–256. DOI: <https://doi.org/10.1037/h0043083>.

Shepard, Roger N (1962a). "The analysis of proximities: multidimensional scaling with an unknown distance function. I." In: *Psychometrika* 27.2, pp. 125–140.

Shepard, Roger N (1962b). "The analysis of proximities: Multidimensional scaling with an unknown distance function. II." In: *Psychometrika* 27.3, pp. 219–246.

- Shepard, Roger N (1986). “Discrimination and generalization in identification and classification: Comment on Nosofsky.” In: 115.1, pp. 58–61. DOI: <https://doi.org/10.1037/0096-3445.115.1.58>.
- Shi, Lei, Naomi H Feldman, and Thomas L Griffiths (2008). “Performing Bayesian inference with exemplar models.” In: *Proceedings of the Annual Meeting of the Cognitive Science Society*. Vol. 30. 30, pp. 745–750.
- Sidman, Murray (1952). “A note on functional relations obtained from group data.” In: *Psychological Bulletin* 49.3, pp. 263–269.
- Silverman, Daniel (2012). “Mikołaj Kruszewski: Theory and Vision (Part Two)†.” In: *Language and Linguistics Compass* 6.5, pp. 296–309. DOI: <https://doi.org/10.1002/lnc3.335>. eprint: <https://compass.onlinelibrary.wiley.com/doi/pdf/10.1002/lnc3.335>. URL: <https://compass.onlinelibrary.wiley.com/doi/abs/10.1002/lnc3.335>.
- Sonderegger, Morgan and Alan C.L. Yu (2010). “A rational account of perceptual compensation for coarticulation.” In: *Proceedings of the 32nd Annual Meeting of the Cognitive Science Society (CogSci10)*, pp. 375–380.
- Stern, Michael C (2021). “Phonological Contrast Drives Phonetic Implementation: Evidence from Category Goodness Ratings.” In: *Proceedings of the 44th Annual Penn Linguistics Conference*.
- Tan, Maryann and T. Florian Jaeger (2025). “Learning to understand an unfamiliar talker: Testing distributional learning as a model of rapid adaptive speech perception.” In: *Cognition* 265, p. 106195. ISSN: 0010-0277. DOI: <https://doi.org/10.1016/j.cognition.2025.106195>. URL: <https://www.sciencedirect.com/science/article/pii/S0010027725001350>.
- Tanner, J. W. (2019). *How to check if a function is convex*. Mathematics Stack Exchange. URL: <https://math.stackexchange.com/q/3325382> (version: 2024-11-13). eprint: <https://math.stackexchange.com/q/3325382>.
- Tenenbaum, Joshua B and Thomas L Griffiths (2001). “Generalization, similarity, and Bayesian inference.” In: *Behavioral and brain sciences* 24.4, pp. 629–640.
- Thurstone, Louis L (1927). “Psychophysical analysis.” In: *The American journal of psychology* 38.3, pp. 368–389.

- Toscano, Joseph C, Bob McMurray, Joel Dennhardt, and Steven J Luck (2010). “Continuous perception and graded categorization: Electrophysiological evidence for a linear relationship between the acoustic signal and perceptual encoding of speech.” In: *Psychological science* 21.10, pp. 1532–1540.
- Treisman, Michel (1999). “There are two types of psychometric function: A theory of cue combination in the processing of complex stimuli with implications for categorical perception.” In: *Journal of Experimental Psychology: General* 128.4, p. 517.
- Van Fraassen, Bas C (1980). *The scientific image*. Oxford University Press.
- Van Linden, Sabine and Jean Vroomen (2007). “Recalibration of phonetic categories by lipread speech versus lexical information.” In: *Journal of Experimental Psychology: Human Perception and Performance* 33.6, p. 1483.
- Van Wassenhove, Virginie, Ken W Grant, and David Poeppel (2007). “Temporal window of integration in auditory-visual speech perception.” In: *Neuropsychologia* 45.3, pp. 598–607.
- Vroomen, Jean, Sabine van Linden, Mirjam Keetels, Béatrice De Gelder, and Paul Bertelson (2004). “Selective adaptation and recalibration of auditory speech by lipread information: dissipation.” In: *Speech Communication* 44.1-4, pp. 55–61.
- Wehner, R and M Müller (1993). “How do ants acquire their celestial ephemeris function?” In: *Naturwissenschaften* 80.7, pp. 331–333.
- Wehner, Rudiger (2003). “Desert ant navigation: how miniature brains solve complex tasks.” In: *Journal of Comparative Physiology A* 189.8, pp. 579–588.
- Weide, Robert L. (1994). “CMU Pronouncing Dictionary.” URL: <http://www.speech.cs.cmu.edu/cgi-bin/cmudict>.
- Wolpert, Daniel M and Zoubin Ghahramani (2005). “Bayes rule in perception, action and cognition.” In: *Science*, pp. 1–4.
- Xie, Xin, T. Florian Jaeger, and Chigusa Kurumada (2023). “What we do (not) know about the mechanisms underlying adaptive speech perception: A computational framework and review.” In: *Cortex* 166, pp. 377–424. ISSN: 0010-9452. DOI: <https://doi.org/10.1016/j.cortex.2023.05.003>. URL: <https://www.sciencedirect.com/science/article/pii/S0010945223001107>.

- 1206 Yang, Charles (2001). “Knowledge and learning in natural language.” Ph.D. Dissertation. Massachusetts
1207 Institute of Technology. URL: <https://dspace.mit.edu/handle/1721.1/86586>.
- 1208 Yang, Charles (2016). *The price of productivity*. Boston, MA, USA: MIT press.
- 1209 Yu, Alan C.L. (2011). “On measuring phonetic precursor robustness: a response to Moreton.” In: *Phonology*
1210 28.3, pp. 491–518.